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Contaminant trap system for a reactor system

Abstract

A contaminant trap system of a reactor system may comprise a baffle plate stack comprising at least one baffle plate comprising an aperture spanning through a baffle plate body of the baffle plate, and a solid body portion; and at least one complementary baffle plate comprising a complementary aperture spanning through a complementary baffle plate body of the complementary baffle plate, and a complementary solid body portion. The at least one baffle plate and the at least one complementary baffle plate may be disposed in a baffle plate order between a first end and a second end of the baffle plate stack in which the baffle plates alternate with the complementary baffle plates, such that no two baffle plates or no two complementary baffle plates are adjacent in the baffle plate order.

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5328810	12/1993	Lowrey et al.	N/A	N/A
5335309	12/1993	Fujii et al.	N/A	N/A
5336327	12/1993	Lee	N/A	N/A
5338362	12/1993	Imahashi	N/A	N/A
5340261	12/1993	Oosawa et al.	N/A	N/A
5346961	12/1993	Shaw et al.	N/A	N/A
5348774	12/1993	Golecki et al.	N/A	N/A
5350480	12/1993	Gray	N/A	N/A
5354433	12/1993	Granneman et al.	N/A	N/A
5354580	12/1993	Goela et al.	N/A	N/A
5356478	12/1993	Chen et al.	N/A	N/A
5356672	12/1993	Schmitt et al.	N/A	N/A
5360269	12/1993	Ogawa et al.	N/A	N/A
5362328	12/1993	Gardiner et al.	N/A	N/A
5364667	12/1993	Rhieu	N/A	N/A
D353452	12/1993	Groenhoff	N/A	N/A
5374315	12/1993	Deboer et al.	N/A	N/A
D354898	12/1994	Nagy	N/A	N/A
5378501	12/1994	Foster et al.	N/A	N/A
5380367	12/1994	Bertone	N/A	N/A
5382311	12/1994	Ishikawa et al.	N/A	N/A
5387265	12/1994	Kakizaki et al.	N/A	N/A
5388945	12/1994	Garric et al.	N/A	N/A
5393577	12/1994	Uesugi et al.	N/A	N/A
5397395	12/1994	Sano et al.	N/A	N/A
5403630	12/1994	Matsui et al.	N/A	N/A
5404082	12/1994	Hernandez et al.	N/A	N/A
5407449	12/1994	Zinger	N/A	N/A
5407867	12/1994	Iwasaki et al.	N/A	N/A
5413813	12/1994	Cruse et al.	N/A	N/A
5414221	12/1994	Gardner	N/A	N/A
5415753	12/1994	Hurwitt et al.	N/A	N/A

5418382	12/1994	Blackwood et al.	N/A	N/A
5421893	12/1994	Perlov	N/A	N/A
5422139	12/1994	Fischer	N/A	N/A
5423942	12/1994	Robbins et al.	N/A	N/A
5426137	12/1994	Allen	N/A	N/A
5427824	12/1994	Inushima et al.	N/A	N/A
5430011	12/1994	Tanaka et al.	N/A	N/A
5431734	12/1994	Chapple-Sokol et al.	N/A	N/A
5443646	12/1994	Yamada et al.	N/A	N/A
5443648	12/1994	Ohkase	N/A	N/A
5443686	12/1994	Jones et al.	N/A	N/A
5444217	12/1994	Moore	N/A	N/A
5447294	12/1994	Sakata et al.	N/A	N/A
5453124	12/1994	Moslehi et al.	N/A	N/A
D363464	12/1994	Fukasawa	N/A	N/A
5456207	12/1994	Gedridge et al.	N/A	N/A
5456757	12/1994	Aruga et al.	N/A	N/A
5461214	12/1994	Peck et al.	N/A	N/A
5462899	12/1994	Ikeda	N/A	N/A
5463176	12/1994	Eckert	N/A	N/A
5464313	12/1994	Ohsawa	N/A	N/A
5474410	12/1994	Ozawa et al.	N/A	N/A
5474612	12/1994	Sato et al.	N/A	N/A
5478429	12/1994	Komino et al.	N/A	N/A
5480488	12/1995	Bittner et al.	N/A	N/A
5480818	12/1995	Matsumoto et al.	N/A	N/A
5482559	12/1995	Imai et al.	N/A	N/A
5484484	12/1995	Yamaga et al.	N/A	N/A
5494439	12/1995	Goldstein et al.	N/A	N/A
5494494	12/1995	Mizuno et al.	N/A	N/A
5496408	12/1995	Motoda et al.	N/A	N/A
5501740	12/1995	Besen et al.	N/A	N/A
5503875	12/1995	Imai et al.	N/A	N/A
5504042	12/1995	Cho et al.	N/A	N/A
5510277	12/1995	Cunningham et al.	N/A	N/A
5512102	12/1995	Yamazaki	N/A	N/A
5514439	12/1995	Sibley	N/A	N/A
5518549	12/1995	Hellwig	N/A	N/A
5518780	12/1995	Tamor et al.	N/A	N/A
5519234	12/1995	Paz de Araujo	N/A	N/A
5520743	12/1995	Takahashi et al.	N/A	N/A
5523616	12/1995	Yasuhide	N/A	N/A
5527111	12/1995	Lysen et al.	N/A	N/A
5527417	12/1995	Iida et al.	N/A	N/A
5531218	12/1995	Krebs	N/A	N/A
5531835	12/1995	Fodor et al.	N/A	N/A
5537311	12/1995	Stevens	N/A	N/A
5540059	12/1995	Yokokawa	N/A	N/A
5540821	12/1995	Tepman	N/A	N/A
5540898	12/1995	Davidson	N/A	N/A

5554557	12/1995	Koh	N/A	N/A
5556275	12/1995	Sakata et al.	N/A	N/A
5558717	12/1995	Zhao et al.	N/A	N/A
5559046	12/1995	Oishi et al.	N/A	N/A
5562383	12/1995	Iwai et al.	N/A	N/A
5562947	12/1995	White et al.	N/A	N/A
5562952	12/1995	Nakahigashi et al.	N/A	N/A
5565038	12/1995	Ashley	N/A	N/A
5569402	12/1995	Meisser et al.	N/A	N/A
5574247	12/1995	Nishitani et al.	N/A	N/A
5576629	12/1995	Turner	N/A	N/A
5577331	12/1995	Suzuki	N/A	N/A
5583736	12/1995	Anderson et al.	N/A	N/A
5584936	12/1995	Pickering et al.	N/A	N/A
5584963	12/1995	Takahashi	N/A	N/A
5586585	12/1995	Bonora et al.	N/A	N/A
5589002	12/1995	Su	N/A	N/A
5589110	12/1995	Motoda et al.	N/A	N/A
5595606	12/1996	Fujikawa et al.	N/A	N/A
5601641	12/1996	Stephens	N/A	N/A
5602060	12/1996	Kobayashi et al.	N/A	N/A
5604410	12/1996	Vollkommer et al.	N/A	N/A
5611448	12/1996	Chen	N/A	N/A
5616264	12/1996	Nishi et al.	N/A	N/A
5616947	12/1996	Tamura	N/A	N/A
5621982	12/1996	Yamashita	N/A	N/A
5632919	12/1996	MacCracken et al.	N/A	N/A
5637153	12/1996	Niino et al.	N/A	N/A
D380527	12/1996	Velez	N/A	N/A
5645646	12/1996	Beinglass et al.	N/A	N/A
5650013	12/1996	Yamazaki	N/A	N/A
5650351	12/1996	Wu	N/A	N/A
5653807	12/1996	Crumbaker	N/A	N/A
5656093	12/1996	Burkhart et al.	N/A	N/A
5661263	12/1996	Salvaggio	N/A	N/A
5662470	12/1996	Huussen et al.	N/A	N/A
5663899	12/1996	Zvonar et al.	N/A	N/A
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5667592	12/1996	Boitnott et al.	N/A	N/A
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5681779	12/1996	Pasch et al.	N/A	N/A
D386076	12/1996	Moore	N/A	N/A
5683517	12/1996	Shan	N/A	N/A
5683561	12/1996	Hollars et al.	N/A	N/A
5685912	12/1996	Nishizaka	N/A	N/A
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5690742	12/1996	Ogata et al.	N/A	N/A
5695567	12/1996	Kordina	N/A	N/A
5697706	12/1996	Ciaravino et al.	N/A	N/A

5698036	12/1996	Ishii et al.	N/A	N/A
5700729	12/1996	Lee et al.	N/A	N/A
5708825	12/1997	Sotomayor	N/A	N/A
5709745	12/1997	Larkin et al.	N/A	N/A
5711811	12/1997	Suntola et al.	N/A	N/A
5716133	12/1997	Hosokawa et al.	N/A	N/A
5718574	12/1997	Shimazu	N/A	N/A
5720927	12/1997	Cripe et al.	N/A	N/A
D392855	12/1997	Pillow	N/A	N/A
5724748	12/1997	Brooks	N/A	N/A
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5730802	12/1997	Ishizumi et al.	N/A	N/A
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5732957	12/1997	Yu	N/A	N/A
5736314	12/1997	Hayes et al.	N/A	N/A
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5754390	12/1997	Sandhu et al.	N/A	N/A
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5766365	12/1997	Umutoy et al.	N/A	N/A
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5769952	12/1997	Komino	N/A	N/A
5775889	12/1997	Kobayashi et al.	N/A	N/A
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5779203	12/1997	Edlinger	N/A	N/A
5781693	12/1997	Ballance et al.	N/A	N/A
5782979	12/1997	Kaneno	N/A	N/A
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5801945	12/1997	Comer	N/A	N/A
5804505	12/1997	Yamada et al.	N/A	N/A
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5813851	12/1997	Nakao	N/A	N/A
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5820366	12/1997	Lee	N/A	N/A
5820685	12/1997	Kurihira et al.	N/A	N/A
5820686	12/1997	Moore	N/A	N/A
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5827420	12/1997	Shirazi et al.	N/A	N/A
5827435	12/1997	Samukawa	N/A	N/A
5827757	12/1997	Robinson et al.	N/A	N/A

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5837058	12/1997	Chen et al.	N/A	N/A
5837320	12/1997	Hampden-Smith et al.	N/A	N/A
5844683	12/1997	Pavloski et al.	N/A	N/A
5846332	12/1997	Zhao et al.	N/A	N/A
5851293	12/1997	Lane et al.	N/A	N/A
5851294	12/1997	Young et al.	N/A	N/A
5851299	12/1997	Cheng et al.	N/A	N/A
5852445	12/1997	Yoshikawa et al.	N/A	N/A
5852879	12/1997	Schumaier	N/A	N/A
5853484	12/1997	Jeong	N/A	N/A
D403949	12/1998	Nakamura	N/A	N/A
D404370	12/1998	Kimura	N/A	N/A
D404372	12/1998	Ishii	N/A	N/A
5855680	12/1998	Soininen et al.	N/A	N/A
5855681	12/1998	Maydan et al.	N/A	N/A
5855687	12/1998	DuBois et al.	N/A	N/A
5855726	12/1998	Soraoka et al.	N/A	N/A
5857777	12/1998	Schuh	N/A	N/A
5861233	12/1998	Sekine et al.	N/A	N/A
5862302	12/1998	Okase	N/A	N/A
5863123	12/1998	Lee et al.	N/A	N/A
5865205	12/1998	Wilmer	N/A	N/A
5866795	12/1998	Wang et al.	N/A	N/A
5871586	12/1998	Crawley et al.	N/A	N/A
5872065	12/1998	Sivaramakrishnan	N/A	N/A
5873942	12/1998	Park	N/A	N/A
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5879128	12/1998	Tietz et al.	N/A	N/A
5879459	12/1998	Gadgil et al.	N/A	N/A
5880980	12/1998	Rothacher et al.	N/A	N/A
5882165	12/1998	Maydan et al.	N/A	N/A
5884640	12/1998	Fishkin et al.	N/A	N/A
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5891251	12/1998	MacLeish et al.	N/A	N/A
5893741	12/1998	Huang	N/A	N/A
5897348	12/1998	Wu	N/A	N/A
5897378	12/1998	Eriguchi	N/A	N/A
5897379	12/1998	Ulrich et al.	N/A	N/A
5897710	12/1998	Sato et al.	N/A	N/A
D409894	12/1998	McClurg	N/A	N/A
5904170	12/1998	Harvey et al.	N/A	N/A
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D411516	12/1998	Imafuku et al.	N/A	N/A
5908672	12/1998	Ryu	N/A	N/A
5915200	12/1998	Tokumasu et al.	N/A	N/A
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5916365	12/1998	Sherman	N/A	N/A
D412270	12/1998	Fredrickson	N/A	N/A
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D412512	12/1998	Boisvert	N/A	N/A
5937142	12/1998	Moslehi et al.	N/A	N/A
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5950925	12/1998	Fukunaga et al.	N/A	N/A
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5961775	12/1998	Fujimura	N/A	N/A
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5979506	12/1998	Aarseth	N/A	N/A
5982931	12/1998	Ishimaru	N/A	N/A
5984391	12/1998	Vanderpot et al.	N/A	N/A
5987480	12/1998	Donohue et al.	N/A	N/A
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5992453	12/1998	Zimmer	N/A	N/A
5997588	12/1998	Goodwin	N/A	N/A
5997768	12/1998	Scully	N/A	N/A
5998870	12/1998	Lee et al.	N/A	N/A
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6001183	12/1998	Gurary et al.	N/A	N/A
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D419652	12/1999	Hall et al.	N/A	N/A
6013553	12/1999	Wallace	N/A	N/A
6013920	12/1999	Gordon et al.	N/A	N/A
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6015465	12/1999	Kholodenko et al.	N/A	N/A
6017779	12/1999	Miyasaka	N/A	N/A
6017818	12/1999	Lu	N/A	N/A
6022180	12/1999	Motoyama et al.	N/A	N/A
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6025117	12/1999	Nakano et al.	N/A	N/A
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6033215	12/1999	Ohsawa	N/A	N/A
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6042652	12/1999	Hyun	N/A	N/A
6044860	12/1999	Neu	N/A	N/A
6045260	12/1999	Schwartz et al.	N/A	N/A

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6050506	12/1999	Guo et al.	N/A	N/A
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6053983	12/1999	Saeki et al.	N/A	N/A
6054013	12/1999	Collins et al.	N/A	N/A
6054678	12/1999	Miyazaki	N/A	N/A
6060691	12/1999	Minami et al.	N/A	N/A
6060721	12/1999	Huang	N/A	N/A
6063196	12/1999	Li et al.	N/A	N/A
6066204	12/1999	Haven	N/A	N/A
6068441	12/1999	Raaijmakers et al.	N/A	N/A
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6090212	12/1999	Mahawili	N/A	N/A
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6093252	12/1999	Wengert et al.	N/A	N/A
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6119710	12/1999	Brown	N/A	N/A
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6127249	12/1999	Hu	N/A	N/A
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6164894	12/1999	Cheng	N/A	N/A
6174809	12/2000	Kang et al.	N/A	N/A
6176929	12/2000	Fukunaga et al.	N/A	N/A
6177688	12/2000	Linthicum et al.	N/A	N/A
6178918	12/2000	Van Os et al.	N/A	N/A
6179955	12/2000	Shin et al.	N/A	N/A
6180979	12/2000	Hofman et al.	N/A	N/A
6182603	12/2000	Shang et al.	N/A	N/A
6183564	12/2000	Reynolds et al.	N/A	N/A
6183565	12/2000	Granneman et al.	N/A	N/A
6187672	12/2000	Zhao et al.	N/A	N/A
6187691	12/2000	Fukuda	N/A	N/A
6189482	12/2000	Zhao et al.	N/A	N/A
6190037	12/2000	Das et al.	N/A	N/A
6190113	12/2000	Bui et al.	N/A	N/A
6190634	12/2000	Lieber et al.	N/A	N/A
6191399	12/2000	Van Bilsen	N/A	N/A
6194037	12/2000	Terasaki et al.	N/A	N/A
6200897	12/2000	Wang et al.	N/A	N/A
6201999	12/2000	Jevtic	N/A	N/A
6203613	12/2000	Gates et al.	N/A	N/A
6203618	12/2000	Hashizume et al.	N/A	N/A
6207932	12/2000	Yoo	N/A	N/A
6207936	12/2000	de Waard	N/A	N/A

6209221	12/2000	Beulens	N/A	N/A
6210485	12/2000	Zhao et al.	N/A	N/A
6212789	12/2000	Kato	N/A	N/A
6213708	12/2000	Allen	N/A	N/A
6214122	12/2000	Thompson	N/A	N/A
6214717	12/2000	Yan et al.	N/A	N/A
6217658	12/2000	Orczyk et al.	N/A	N/A
6217662	12/2000	Kong et al.	N/A	N/A
6218288	12/2000	Li et al.	N/A	N/A
6224679	12/2000	Sasaki et al.	N/A	N/A
6225020	12/2000	Jung et al.	N/A	N/A
6225602	12/2000	Buiijze et al.	N/A	N/A
6225745	12/2000	Srivastava	N/A	N/A
6230650	12/2000	Yamazaki	N/A	N/A
6231290	12/2000	Kikuchi et al.	N/A	N/A
6235121	12/2000	Honma et al.	N/A	N/A
6235858	12/2000	Swarup et al.	N/A	N/A
6238734	12/2000	Senzaki et al.	N/A	N/A
6239402	12/2000	Araki et al.	N/A	N/A
6239715	12/2000	Belton	N/A	N/A
6240875	12/2000	Wijck et al.	N/A	N/A
6241822	12/2000	Ide	N/A	N/A
6242359	12/2000	Misra	N/A	N/A
6243654	12/2000	Johnson et al.	N/A	N/A
6245647	12/2000	Akiyama et al.	N/A	N/A
6245665	12/2000	Yokoyama	N/A	N/A
6247245	12/2000	Ishii	N/A	N/A
6250250	12/2000	Maishev et al.	N/A	N/A
6255221	12/2000	Hudson et al.	N/A	N/A
6257758	12/2000	Culbertson	N/A	N/A
6261648	12/2000	Akiba et al.	N/A	N/A
6264467	12/2000	Lue et al.	N/A	N/A
6265113	12/2000	Rosasco et al.	N/A	N/A
6265311	12/2000	Hautala et al.	N/A	N/A
6270572	12/2000	Kim et al.	N/A	N/A
6271148	12/2000	Kao	N/A	N/A
6271320	12/2000	Keller et al.	N/A	N/A
6274496	12/2000	Leusink et al.	N/A	N/A
6274878	12/2000	Li et al.	N/A	N/A
6281098	12/2000	Wang	N/A	N/A
6281141	12/2000	Das et al.	N/A	N/A
6283692	12/2000	Perlov et al.	N/A	N/A
6284050	12/2000	Shi et al.	N/A	N/A
6284149	12/2000	Li et al.	N/A	N/A
6287965	12/2000	Kang et al.	N/A	N/A
6287988	12/2000	Nagamine et al.	N/A	N/A
6293700	12/2000	Lund et al.	N/A	N/A
D449873	12/2000	Bronson	N/A	N/A
6296710	12/2000	Allen et al.	N/A	N/A
6296711	12/2000	Loan et al.	N/A	N/A

6296909	12/2000	Spitsberg	N/A	N/A
6297539	12/2000	Ma et al.	N/A	N/A
6299133	12/2000	Waragai et al.	N/A	N/A
6302964	12/2000	Umotoy et al.	N/A	N/A
6303523	12/2000	Cheung	N/A	N/A
6305898	12/2000	Yamagishi et al.	N/A	N/A
6311016	12/2000	Yanagawa et al.	N/A	N/A
6312525	12/2000	Bright et al.	N/A	N/A
6315512	12/2000	Tabrizi et al.	N/A	N/A
6316162	12/2000	Jung et al.	N/A	N/A
6316371	12/2000	Oosterlaken et al.	N/A	N/A
6320320	12/2000	Bailey et al.	N/A	N/A
6321680	12/2000	Cook et al.	N/A	N/A
6321780	12/2000	Iwabuchi	N/A	N/A
D451893	12/2000	Robson	N/A	N/A
D452220	12/2000	Robson	N/A	N/A
6325858	12/2000	Wengert	N/A	N/A
6326322	12/2000	Kim et al.	N/A	N/A
6326597	12/2000	Lubomirsky et al.	N/A	N/A
6328561	12/2000	Hasper et al.	N/A	N/A
6328864	12/2000	Ishizawa et al.	N/A	N/A
6329297	12/2000	Balish	N/A	N/A
6333275	12/2000	Mayer et al.	N/A	N/A
6335049	12/2001	Basceri	N/A	N/A
6335240	12/2001	Kim et al.	N/A	N/A
6335293	12/2001	Luo et al.	N/A	N/A
6342427	12/2001	Choi et al.	N/A	N/A
6343239	12/2001	Toda et al.	N/A	N/A
6344084	12/2001	Koinuma et al.	N/A	N/A
6344232	12/2001	Jones et al.	N/A	N/A
6346419	12/2001	Ryerson et al.	N/A	N/A
6347636	12/2001	Xia	N/A	N/A
6350391	12/2001	Livshits et al.	N/A	N/A
6352049	12/2001	Yin et al.	N/A	N/A
6352945	12/2001	Matsuki	N/A	N/A
6357984	12/2001	Zinger et al.	N/A	N/A
6363294	12/2001	Coronel et al.	N/A	N/A
D455024	12/2001	Mimick et al.	N/A	N/A
6367410	12/2001	Leahey et al.	N/A	N/A
6368773	12/2001	Jung et al.	N/A	N/A
6368987	12/2001	Kopacz et al.	N/A	N/A
6370796	12/2001	Zucker	N/A	N/A
6372583	12/2001	Tyagi	N/A	N/A
6374831	12/2001	Chandran	N/A	N/A
6375312	12/2001	Ikeda et al.	N/A	N/A
6375749	12/2001	Boydston et al.	N/A	N/A
6375750	12/2001	Van Os et al.	N/A	N/A
6379466	12/2001	Sahin et al.	N/A	N/A
D457609	12/2001	Piano	N/A	N/A
6383300	12/2001	Saito et al.	N/A	N/A

6383566	12/2001	Zagdoun	N/A	N/A
6383955	12/2001	Matsuki	N/A	N/A
6387207	12/2001	Janakiraman	N/A	N/A
6387823	12/2001	Sonderman et al.	N/A	N/A
6387827	12/2001	Mertens et al.	N/A	N/A
6390753	12/2001	de Ridder et al.	N/A	N/A
6390754	12/2001	Yamaga et al.	N/A	N/A
6391803	12/2001	Kim et al.	N/A	N/A
6395650	12/2001	Callegari et al.	N/A	N/A
6398184	12/2001	Sowada et al.	N/A	N/A
6399022	12/2001	Schuler et al.	N/A	N/A
6402806	12/2001	Schmitt et al.	N/A	N/A
6410433	12/2001	Hautala et al.	N/A	N/A
6410459	12/2001	Blalock et al.	N/A	N/A
6410463	12/2001	Matsuki	N/A	N/A
6413321	12/2001	Kim et al.	N/A	N/A
6413583	12/2001	Moghadam et al.	N/A	N/A
6420279	12/2001	Ono et al.	N/A	N/A
6423949	12/2001	Chen et al.	N/A	N/A
D461233	12/2001	Whalen	N/A	N/A
D461882	12/2001	Piano	N/A	N/A
6428859	12/2001	Chiang et al.	N/A	N/A
6429465	12/2001	Yagi et al.	N/A	N/A
6432206	12/2001	Tolt	N/A	N/A
6432255	12/2001	Sun et al.	N/A	N/A
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6432849	12/2001	Endo et al.	N/A	N/A
6433298	12/2001	Ishii	N/A	N/A
6435798	12/2001	Satoh	N/A	N/A
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6437444	12/2001	Andideh	N/A	N/A
6438502	12/2001	Awtrey	N/A	N/A
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6447232	12/2001	Davis et al.	N/A	N/A
6447651	12/2001	Ishikawa et al.	N/A	N/A
6447937	12/2001	Murakawa et al.	N/A	N/A
6448192	12/2001	Kaushik	N/A	N/A
6450117	12/2001	Murugesh et al.	N/A	N/A
6450757	12/2001	Saeki	N/A	N/A
6451713	12/2001	Tay et al.	N/A	N/A
6452017	12/2001	Uhlenbrock et al.	N/A	N/A
6454860	12/2001	Metzner et al.	N/A	N/A
6454909	12/2001	Matsuse et al.	N/A	N/A
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6455225	12/2001	Kong et al.	N/A	N/A

6455445	12/2001	Matsuki	N/A	N/A
6460482	12/2001	Kuibira et al.	N/A	N/A
6461435	12/2001	Littau et al.	N/A	N/A
6461436	12/2001	Campbell et al.	N/A	N/A
6461439	12/2001	Granneman et al.	N/A	N/A
6462310	12/2001	Ratliff et al.	N/A	N/A
6464825	12/2001	Shinozaki	N/A	N/A
6468924	12/2001	Lee	N/A	N/A
6471779	12/2001	Nishio et al.	N/A	N/A
6472266	12/2001	Yu et al.	N/A	N/A
6474987	12/2001	Huang et al.	N/A	N/A
6475276	12/2001	Elers et al.	N/A	N/A
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6481945	12/2001	Hasper et al.	N/A	N/A
6482331	12/2001	Lu et al.	N/A	N/A
6482663	12/2001	Buckland	N/A	N/A
6483989	12/2001	Okada et al.	N/A	N/A
6488774	12/2001	Horie et al.	N/A	N/A
6490493	12/2001	Dharnipragada	N/A	N/A
6492625	12/2001	Boguslavskiy et al.	N/A	N/A
6494065	12/2001	Babbitt	N/A	N/A
6494998	12/2001	Brcka	N/A	N/A
6496819	12/2001	Bello et al.	N/A	N/A
6497734	12/2001	Barber et al.	N/A	N/A
6497767	12/2001	Okase et al.	N/A	N/A
6498091	12/2001	Chen et al.	N/A	N/A
6499533	12/2001	Yamada	N/A	N/A
6500487	12/2001	Holst et al.	N/A	N/A
6502530	12/2002	Turlot et al.	N/A	N/A
6503079	12/2002	Kogano et al.	N/A	N/A
6503330	12/2002	Sneh et al.	N/A	N/A
6503365	12/2002	Kim et al.	N/A	N/A
6503562	12/2002	Saito et al.	N/A	N/A
6503826	12/2002	Oda	N/A	N/A
6506009	12/2002	Nulman et al.	N/A	N/A
6506253	12/2002	Sakuma	N/A	N/A
6507410	12/2002	Robertson et al.	N/A	N/A
6511539	12/2002	Raaijmakers	N/A	N/A
6514313	12/2002	Spiegelman	N/A	N/A
6514666	12/2002	Choi et al.	N/A	N/A
6521295	12/2002	Remington	N/A	N/A
6521547	12/2002	Chang et al.	N/A	N/A
6527884	12/2002	Takakuwa et al.	N/A	N/A
6528171	12/2002	Endler et al.	N/A	N/A
6528430	12/2002	Kwan	N/A	N/A
6528752	12/2002	Ishii et al.	N/A	N/A
6528767	12/2002	Bagley et al.	N/A	N/A
6530994	12/2002	Mahawili	N/A	N/A

6531193	12/2002	Fonash et al.	N/A	N/A
6531412	12/2002	Conti et al.	N/A	N/A
6534133	12/2002	Kaloyeros et al.	N/A	N/A
6534395	12/2002	Werkhoven et al.	N/A	N/A
6536950	12/2002	Green	N/A	N/A
6539891	12/2002	Kang et al.	N/A	N/A
6540469	12/2002	Matsunaga et al.	N/A	N/A
6540838	12/2002	Sneh et al.	N/A	N/A
6544906	12/2002	Rotondaro et al.	N/A	N/A
6547844	12/2002	Rikyuu et al.	N/A	N/A
6552209	12/2002	Lei et al.	N/A	N/A
6554879	12/2002	Nomura	N/A	N/A
6558517	12/2002	Basceri	N/A	N/A
6558755	12/2002	Berry et al.	N/A	N/A
6559026	12/2002	Rossman et al.	N/A	N/A
6562094	12/2002	Denker et al.	N/A	N/A
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6566278	12/2002	Harvey et al.	N/A	N/A
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6573030	12/2002	Fairbairn et al.	N/A	N/A
6574644	12/2002	Hsu et al.	N/A	N/A
6576062	12/2002	Matsuse	N/A	N/A
6576064	12/2002	Griffiths et al.	N/A	N/A
6576300	12/2002	Berry et al.	N/A	N/A
6576564	12/2002	Agarwal	N/A	N/A
6578589	12/2002	Mayusumi	N/A	N/A
6579833	12/2002	McNallan et al.	N/A	N/A
6580050	12/2002	Miller et al.	N/A	N/A
6582174	12/2002	Hayashi	N/A	N/A
6583048	12/2002	Vincent et al.	N/A	N/A
6583572	12/2002	Veltrop et al.	N/A	N/A
6587108	12/2002	Guerlain et al.	N/A	N/A
6589352	12/2002	Yudovsky et al.	N/A	N/A
6589707	12/2002	Lee et al.	N/A	N/A
6589868	12/2002	Rossman	N/A	N/A
6590251	12/2002	Kang et al.	N/A	N/A
6594550	12/2002	Okrah	N/A	N/A
6596398	12/2002	Russo et al.	N/A	N/A
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6602806	12/2002	Xia et al.	N/A	N/A
6607602	12/2002	Granneman et al.	N/A	N/A
6607868	12/2002	Choi	N/A	N/A
6607948	12/2002	Sugiyama et al.	N/A	N/A
6608745	12/2002	Tsuruta et al.	N/A	N/A
6610375	12/2002	Akiba et al.	N/A	N/A
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6616986	12/2002	Sherman	N/A	N/A
6617253	12/2002	Chu et al.	N/A	N/A

6620251	12/2002	Kitano	N/A	N/A
6623799	12/2002	Lee et al.	N/A	N/A
6624064	12/2002	Sahin	N/A	N/A
6627268	12/2002	Fair et al.	N/A	N/A
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6630030	12/2002	Suntola et al.	N/A	N/A
6632478	12/2002	Gaillard et al.	N/A	N/A
6633364	12/2002	Hayashi	N/A	N/A
6635115	12/2002	Fairbairn et al.	N/A	N/A
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6638839	12/2002	Deng et al.	N/A	N/A
6640145	12/2002	Hoffberg et al.	N/A	N/A
6645304	12/2002	Yamaguchi	N/A	N/A
6648974	12/2002	Ogliari et al.	N/A	N/A
6649921	12/2002	Cekic et al.	N/A	N/A
6652924	12/2002	Sherman	N/A	N/A
6656281	12/2002	Ueda	N/A	N/A
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6658933	12/2002	Allegre et al.	N/A	N/A
6659111	12/2002	Mouri et al.	N/A	N/A
6660662	12/2002	Ishikawa et al.	N/A	N/A
6662817	12/2002	Yamagishi	N/A	N/A
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6673196	12/2003	Oyabu	N/A	N/A
6676290	12/2003	Lu	N/A	N/A
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D486891	12/2003	Cronce	N/A	N/A
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6688784	12/2003	Templeton	N/A	N/A
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6692575	12/2003	Omstead et al.	N/A	N/A
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6696367	12/2003	Aggarwal	N/A	N/A
6699003	12/2003	Saeki	N/A	N/A
6699399	12/2003	Qian et al.	N/A	N/A
6700089	12/2003	Hirooka	N/A	N/A
6709989	12/2003	Ramdani et al.	N/A	N/A
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6712949	12/2003	Gopal	N/A	N/A
6713824	12/2003	Mikata	N/A	N/A

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6716571	12/2003	Gabriel	N/A	N/A
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6745095	12/2003	Ben-Dov	N/A	N/A
6746240	12/2003	de Ridder et al.	N/A	N/A
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6760981	12/2003	Leap	N/A	N/A
6766545	12/2003	Hodges	N/A	N/A
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D494552	12/2003	Tezuka et al.	N/A	N/A
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6784108	12/2003	Donohoe et al.	N/A	N/A
D496008	12/2003	Takahashi et al.	N/A	N/A
6786997	12/2003	Yamazaki	N/A	N/A
D497536	12/2003	Fujiwara	N/A	N/A
D497977	12/2003	Engelbrektsson	N/A	N/A
6811960	12/2003	Lee et al.	N/A	N/A
6812157	12/2003	Gadgil	N/A	N/A
6815350	12/2003	Kim et al.	N/A	N/A
6815352	12/2003	Tamura et al.	N/A	N/A
6818566	12/2003	Leeson et al.	N/A	N/A
6818864	12/2003	Ptak	N/A	N/A
6820570	12/2003	Kilpela et al.	N/A	N/A
6821889	12/2003	Elers et al.	N/A	N/A
6821910	12/2003	Adomaitis et al.	N/A	N/A
6824665	12/2003	Shelnut et al.	N/A	N/A

6825106	12/2003	Gao et al.	N/A	N/A
6825134	12/2003	Law et al.	N/A	N/A
D499620	12/2003	Horner-Richardson et al.	N/A	N/A
6827789	12/2003	Lee et al.	N/A	N/A
6828235	12/2003	Takano	N/A	N/A
6831004	12/2003	Byun	N/A	N/A
6833024	12/2003	Holst et al.	N/A	N/A
6835039	12/2003	Van Den Berg	N/A	N/A
6838122	12/2004	Basceri et al.	N/A	N/A
6841201	12/2004	Shanov et al.	N/A	N/A
6843202	12/2004	Kusuda	N/A	N/A
6843858	12/2004	Rossman	N/A	N/A
6846146	12/2004	Inui	N/A	N/A
6846515	12/2004	Vrtis	N/A	N/A
6846742	12/2004	Rossman	N/A	N/A
6847014	12/2004	Benjamin et al.	N/A	N/A
6849241	12/2004	Dauelsberg et al.	N/A	N/A
6849857	12/2004	Ichiki et al.	N/A	N/A
6854580	12/2004	Braford	N/A	N/A
6858524	12/2004	Haukka et al.	N/A	N/A
6858547	12/2004	Metzner	N/A	N/A
6861642	12/2004	Ichiki et al.	N/A	N/A
6863019	12/2004	Shamouilian	N/A	N/A
6863281	12/2004	Endou et al.	N/A	N/A
6864041	12/2004	Brown	N/A	N/A
6867086	12/2004	Chen et al.	N/A	N/A
6867153	12/2004	Tokunaga	N/A	N/A
6867859	12/2004	Powell	N/A	N/A
6872258	12/2004	Park et al.	N/A	N/A
6872259	12/2004	Strang	N/A	N/A
D504142	12/2004	Horner-Richardson et al.	N/A	N/A
6874247	12/2004	Hsu	N/A	N/A
6874480	12/2004	Ismailov	N/A	N/A
6875477	12/2004	Trickett et al.	N/A	N/A
6875677	12/2004	Conley, Jr. et al.	N/A	N/A
6876017	12/2004	Goodner	N/A	N/A
6876191	12/2004	de Ridder	N/A	N/A
6878206	12/2004	Tzu et al.	N/A	N/A
6878402	12/2004	Chiang et al.	N/A	N/A
6883733	12/2004	Lind	N/A	N/A
6884066	12/2004	Nguyen et al.	N/A	N/A
6884295	12/2004	Ishii	N/A	N/A
6884319	12/2004	Kim	N/A	N/A
6884475	12/2004	Basceri	N/A	N/A
D505590	12/2004	Greiner	N/A	N/A
6889211	12/2004	Yoshiura et al.	N/A	N/A
6889864	12/2004	Lindfors et al.	N/A	N/A
6890596	12/2004	Sarigiannis et al.	N/A	N/A

6895158	12/2004	Alyward et al.	N/A	N/A
6899507	12/2004	Yamagishi et al.	N/A	N/A
6902395	12/2004	Oosterlaken et al.	N/A	N/A
6902647	12/2004	Hasper	N/A	N/A
6909839	12/2004	Wang et al.	N/A	N/A
6911092	12/2004	Sneh	N/A	N/A
6913152	12/2004	Zuk	N/A	N/A
6913796	12/2004	Albano et al.	N/A	N/A
6916398	12/2004	Chen et al.	N/A	N/A
6916559	12/2004	Murakawa et al.	N/A	N/A
6917755	12/2004	Nguyen et al.	N/A	N/A
6924078	12/2004	Lee et al.	N/A	N/A
6928890	12/2004	Gehner et al.	N/A	N/A
6929699	12/2004	Whitesell	N/A	N/A
6929700	12/2004	Tan et al.	N/A	N/A
6930041	12/2004	Agarwal	N/A	N/A
6930059	12/2004	Conley, Jr. et al.	N/A	N/A
6935269	12/2004	Lee et al.	N/A	N/A
6939579	12/2004	Bondestam et al.	N/A	N/A
6939817	12/2004	Sandhu et al.	N/A	N/A
6942753	12/2004	Choi et al.	N/A	N/A
6949204	12/2004	Lenz et al.	N/A	N/A
6951587	12/2004	Narushima	N/A	N/A
6952656	12/2004	Cordova et al.	N/A	N/A
6953609	12/2004	Carollo	N/A	N/A
6955741	12/2004	Yamagishi	N/A	N/A
6955836	12/2004	Kumagai et al.	N/A	N/A
6955928	12/2004	Brennan	N/A	N/A
6963052	12/2004	Kuibira et al.	N/A	N/A
6966936	12/2004	Yamasaki et al.	N/A	N/A
6972055	12/2004	Sferlazzo	N/A	N/A
6972478	12/2004	Waite et al.	N/A	N/A
6974781	12/2004	Timmermans et al.	N/A	N/A
6975921	12/2004	Verhaar	N/A	N/A
6976822	12/2004	Woodruff	N/A	N/A
RE38937	12/2005	Nakamura	N/A	N/A
6981832	12/2005	Zinger et al.	N/A	N/A
6982046	12/2005	Srivastava et al.	N/A	N/A
6982103	12/2005	Basceri et al.	N/A	N/A
6984591	12/2005	Buchanan et al.	N/A	N/A
6984595	12/2005	Yamazaki	N/A	N/A
6985788	12/2005	Haanstra et al.	N/A	N/A
6986914	12/2005	Elers et al.	N/A	N/A
6987155	12/2005	Roh et al.	N/A	N/A
6990430	12/2005	Hosek	N/A	N/A
7005227	12/2005	Yueh et al.	N/A	N/A
7005391	12/2005	Min	N/A	N/A
7008879	12/2005	Lee et al.	N/A	N/A
7010580	12/2005	Fu et al.	N/A	N/A
7017514	12/2005	Shepherd et al.	N/A	N/A

7018941	12/2005	Cui et al.	N/A	N/A
7021330	12/2005	Maula et al.	N/A	N/A
7021881	12/2005	Yamagishi	N/A	N/A
7036453	12/2005	Ishikawa et al.	N/A	N/A
7041609	12/2005	Vaartstra	N/A	N/A
7045430	12/2005	Ahn et al.	N/A	N/A
7049226	12/2005	Chung et al.	N/A	N/A
7049247	12/2005	Gates et al.	N/A	N/A
7052584	12/2005	Basceri	N/A	N/A
7053009	12/2005	Conley, Jr. et al.	N/A	N/A
7055263	12/2005	Wu et al.	N/A	N/A
7055875	12/2005	Bonora	N/A	N/A
7062161	12/2005	Kusuda et al.	N/A	N/A
D524600	12/2005	Austin et al.	N/A	N/A
D525127	12/2005	Cogley et al.	N/A	N/A
7070178	12/2005	Van Der Toorn et al.	N/A	N/A
7071051	12/2005	Jeon et al.	N/A	N/A
7073834	12/2005	Matsumoto et al.	N/A	N/A
7074690	12/2005	Gauri et al.	N/A	N/A
7077614	12/2005	Hasper et al.	N/A	N/A
7080545	12/2005	Dimeo et al.	N/A	N/A
7084060	12/2005	Furukawa	N/A	N/A
7084079	12/2005	Conti et al.	N/A	N/A
7085623	12/2005	Siegers	N/A	N/A
7086347	12/2005	Howald et al.	N/A	N/A
7088003	12/2005	Gates et al.	N/A	N/A
7090394	12/2005	Hashikura et al.	N/A	N/A
7092287	12/2005	Beulens et al.	N/A	N/A
7098149	12/2005	Lukas	N/A	N/A
7098150	12/2005	Misra et al.	N/A	N/A
7100459	12/2005	Gehner et al.	N/A	N/A
7101763	12/2005	Anderson et al.	N/A	N/A
7108753	12/2005	Wood	N/A	N/A
7109098	12/2005	Ramaswamy et al.	N/A	N/A
7109114	12/2005	Chen et al.	N/A	N/A
7111232	12/2005	Bascom	N/A	N/A
7115305	12/2005	Bronikowski et al.	N/A	N/A
7115838	12/2005	Kurara et al.	N/A	N/A
7122085	12/2005	Shero et al.	N/A	N/A
7122222	12/2005	Xiao et al.	N/A	N/A
7122844	12/2005	Nakamura et al.	N/A	N/A
7129165	12/2005	Basol et al.	N/A	N/A
7132360	12/2005	Schaeffer et al.	N/A	N/A
7135421	12/2005	Ahn et al.	N/A	N/A
7140558	12/2005	McCracken et al.	N/A	N/A
7141499	12/2005	Raaijmakers	N/A	N/A
7143897	12/2005	Guzman et al.	N/A	N/A
7144806	12/2005	Fair et al.	N/A	N/A
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7147766	12/2005	Uzoh et al.	N/A	N/A

7153542	12/2006	Nguyen et al.	N/A	N/A
D535673	12/2006	Conway et al.	N/A	N/A
7156380	12/2006	Soininen	N/A	N/A
7163393	12/2006	Adachi et al.	N/A	N/A
7163721	12/2006	Zhang et al.	N/A	N/A
7163900	12/2006	Weber	N/A	N/A
7168852	12/2006	Linnarsson	N/A	N/A
7172497	12/2006	Basol et al.	N/A	N/A
7173216	12/2006	Ptak	N/A	N/A
7183229	12/2006	Yamanaka	N/A	N/A
7186648	12/2006	Rozbicki	N/A	N/A
7192824	12/2006	Ahn et al.	N/A	N/A
7192892	12/2006	Ahn et al.	N/A	N/A
7195479	12/2006	Beatty et al.	N/A	N/A
7195693	12/2006	Cowans	N/A	N/A
D541125	12/2006	Gaudron	N/A	N/A
7198447	12/2006	Morimitsu et al.	N/A	N/A
7199513	12/2006	Huber et al.	N/A	N/A
7201943	12/2006	Park et al.	N/A	N/A
7202148	12/2006	Chen et al.	N/A	N/A
7202512	12/2006	Chen et al.	N/A	N/A
7204886	12/2006	Chen et al.	N/A	N/A
7204887	12/2006	Kawamura et al.	N/A	N/A
7205246	12/2006	MacNeil et al.	N/A	N/A
7205247	12/2006	Lee et al.	N/A	N/A
7207763	12/2006	Lee	N/A	N/A
7208198	12/2006	Basceri et al.	N/A	N/A
7208389	12/2006	Tipton et al.	N/A	N/A
7208413	12/2006	Byun et al.	N/A	N/A
7210925	12/2006	Adachi	N/A	N/A
7211524	12/2006	Ryu et al.	N/A	N/A
7211525	12/2006	Shanker	N/A	N/A
7214630	12/2006	Varadarajan et al.	N/A	N/A
7217617	12/2006	Basceri	N/A	N/A
7223014	12/2006	Lojen	N/A	N/A
7229502	12/2006	Wang et al.	N/A	N/A
7234476	12/2006	Arai	N/A	N/A
7235137	12/2006	Kitayama et al.	N/A	N/A
7235482	12/2006	Wu	N/A	N/A
7235501	12/2006	Ahn et al.	N/A	N/A
7238596	12/2006	Kouvetakis et al.	N/A	N/A
7238616	12/2006	Agarwal	N/A	N/A
7238653	12/2006	Lee et al.	N/A	N/A
D549815	12/2006	Murphy	N/A	N/A
7256375	12/2006	Oosterlaken	N/A	N/A
7265061	12/2006	Cho et al.	N/A	N/A
7274867	12/2006	Peukert	N/A	N/A
D553104	12/2006	Oohashi et al.	N/A	N/A
7279256	12/2006	Son	N/A	N/A
7290813	12/2006	Bonora	N/A	N/A

7294581	12/2006	Haverkort et al.	N/A	N/A
7296460	12/2006	Dimeo et al.	N/A	N/A
7297641	12/2006	Todd et al.	N/A	N/A
7298009	12/2006	Yan et al.	N/A	N/A
7301623	12/2006	Madsen et al.	N/A	N/A
D556704	12/2006	Nakamura et al.	N/A	N/A
D557226	12/2006	Uchino et al.	N/A	N/A
D558021	12/2006	Lawrence	N/A	N/A
7307028	12/2006	Goto et al.	N/A	N/A
7307178	12/2006	Kiyomori et al.	N/A	N/A
7311977	12/2006	Yokota et al.	N/A	N/A
7312148	12/2006	Ramaswamy et al.	N/A	N/A
7312162	12/2006	Ramaswamy et al.	N/A	N/A
7312494	12/2006	Ahn et al.	N/A	N/A
D559993	12/2007	Nagakubo et al.	N/A	N/A
D559994	12/2007	Nagakubo et al.	N/A	N/A
7320544	12/2007	Hsieh	N/A	N/A
7323401	12/2007	Ramaswamy et al.	N/A	N/A
D562357	12/2007	Hardy	N/A	N/A
7326656	12/2007	Brask et al.	N/A	N/A
7326657	12/2007	Xia et al.	N/A	N/A
7327948	12/2007	Shrinivasan	N/A	N/A
7329947	12/2007	Adachi et al.	N/A	N/A
7335611	12/2007	Ramaswamy et al.	N/A	N/A
7351057	12/2007	Berenbak et al.	N/A	N/A
7354482	12/2007	Konishi et al.	N/A	N/A
7354847	12/2007	Chan et al.	N/A	N/A
7354873	12/2007	Fukazawa et al.	N/A	N/A
7356762	12/2007	van Driel	N/A	N/A
7357138	12/2007	Ji et al.	N/A	N/A
7361447	12/2007	Jung	N/A	N/A
7375035	12/2007	Heden et al.	N/A	N/A
7376520	12/2007	Wong	N/A	N/A
7378618	12/2007	Sorabji et al.	N/A	N/A
7379785	12/2007	Higashi et al.	N/A	N/A
D571383	12/2007	Ota et al.	N/A	N/A
D571831	12/2007	Ota et al.	N/A	N/A
7381644	12/2007	Soubramonium et al.	N/A	N/A
7387685	12/2007	Choi et al.	N/A	N/A
7393207	12/2007	Imai	N/A	N/A
7393418	12/2007	Yokogawa	N/A	N/A
7393736	12/2007	Ahn et al.	N/A	N/A
7393765	12/2007	Hanawa et al.	N/A	N/A
7396491	12/2007	Marking et al.	N/A	N/A
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7405454	12/2007	Ahn et al.	N/A	N/A
D575713	12/2007	Ratcliffe	N/A	N/A

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7410290	12/2007	Tanaka	N/A	N/A
7410666	12/2007	Elers	N/A	N/A
7411352	12/2007	Madocks	N/A	N/A
7414281	12/2007	Fastow	N/A	N/A
D576001	12/2007	Brunderman	N/A	N/A
7422635	12/2007	Zheng et al.	N/A	N/A
7422636	12/2007	Ishizaka	N/A	N/A
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7422775	12/2007	Ramaswamy et al.	N/A	N/A
7425224	12/2007	Nguyen	N/A	N/A
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7437060	12/2007	Wang et al.	N/A	N/A
7442275	12/2007	Cowans	N/A	N/A
7456429	12/2007	Levy	N/A	N/A
D583395	12/2007	Ueda	N/A	N/A
7467632	12/2007	Lee et al.	N/A	N/A
7473655	12/2008	Wang et al.	N/A	N/A
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7476291	12/2008	Wang et al.	N/A	N/A
7479198	12/2008	Guffrey	N/A	N/A
7482247	12/2008	Papasouliotis	N/A	N/A
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D585968	12/2008	Elkins et al.	N/A	N/A
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D590933	12/2008	Vansell	N/A	N/A
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D593585	12/2008	Ota et al.	N/A	N/A
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D596476	12/2008	Welch	N/A	N/A
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D600223	12/2008	Aggarwal	N/A	N/A
7582555	12/2008	Lang	N/A	N/A
7582575	12/2008	Fukazawa et al.	N/A	N/A
7589003	12/2008	Kouvetakis et al.	N/A	N/A
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D602575	12/2008	Breda	N/A	N/A
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7611640	12/2008	Howald et al.	N/A	N/A
7611751	12/2008	Elers	N/A	N/A
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7618226	12/2008	Takizawa	N/A	N/A
7621672	12/2008	Ripley	N/A	N/A
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D606952	12/2008	Lee et al.	N/A	N/A
7625820	12/2008	Papasouliotis	N/A	N/A
7629277	12/2008	Ghatnagar	N/A	N/A
7632549	12/2008	Goundar	N/A	N/A
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7651583	12/2009	Kent et al.	N/A	N/A
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7651959	12/2009	Fukazawa et al.	N/A	N/A
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D609655	12/2009	Sugimoto	N/A	N/A
7661299	12/2009	Kusunoki	N/A	N/A
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7678197	12/2009	Maki	N/A	N/A
7678715	12/2009	Mungekar et al.	N/A	N/A
7682454	12/2009	Sneh	N/A	N/A
7682657	12/2009	Sherman	N/A	N/A
D613829	12/2009	Griffin et al.	N/A	N/A

D614153	12/2009	Fondurulia et al.	N/A	N/A
D614258	12/2009	Kojima	N/A	N/A
D614267	12/2009	Breda	N/A	N/A
D614268	12/2009	Breda	N/A	N/A
D614593	12/2009	Lee et al.	N/A	N/A
7690881	12/2009	Yamagishi	N/A	N/A
7691205	12/2009	Ikedo	N/A	N/A
7692171	12/2009	Kaszuba et al.	N/A	N/A
7695808	12/2009	Tuma	N/A	N/A
D616390	12/2009	Sato	N/A	N/A
D616394	12/2009	Sato	N/A	N/A
7712435	12/2009	Yoshizaki et al.	N/A	N/A
7713874	12/2009	Milligan	N/A	N/A
7716993	12/2009	Ozawa et al.	N/A	N/A
7718930	12/2009	Kawasaki et al.	N/A	N/A
7720560	12/2009	Menser et al.	N/A	N/A
7723648	12/2009	Tsukamoto et al.	N/A	N/A
7725012	12/2009	Aggarwal et al.	N/A	N/A
7727296	12/2009	Tojo et al.	N/A	N/A
7727864	12/2009	Elers	N/A	N/A
7727880	12/2009	Chattopadhyay et al.	N/A	N/A
7732343	12/2009	Niroomand et al.	N/A	N/A
7736437	12/2009	Cadwell et al.	N/A	N/A
7736528	12/2009	Okita et al.	N/A	N/A
7736600	12/2009	Clark et al.	N/A	N/A
7737035	12/2009	Lind et al.	N/A	N/A
7740437	12/2009	de Ridder et al.	N/A	N/A
7740705	12/2009	Li	N/A	N/A
7745346	12/2009	Hausmann et al.	N/A	N/A
7748760	12/2009	Kushida	N/A	N/A
7749563	12/2009	Zheng et al.	N/A	N/A
7753584	12/2009	Gambino et al.	N/A	N/A
7754621	12/2009	Putjkonen	N/A	N/A
7758698	12/2009	Bang et al.	N/A	N/A
7763869	12/2009	Matsushita et al.	N/A	N/A
7767262	12/2009	Clark	N/A	N/A
7771796	12/2009	Kohno et al.	N/A	N/A
7780440	12/2009	Shibagaki et al.	N/A	N/A
7780789	12/2009	Wu et al.	N/A	N/A
7781352	12/2009	Fukazawa et al.	N/A	N/A
7789559	12/2009	Waser et al.	N/A	N/A
7789965	12/2009	Matsushita et al.	N/A	N/A
7790633	12/2009	Tarafdar et al.	N/A	N/A
7794546	12/2009	Li	N/A	N/A
7795160	12/2009	Wang et al.	N/A	N/A
7798096	12/2009	Mahajani et al.	N/A	N/A
7799300	12/2009	Lindfors et al.	N/A	N/A
7799706	12/2009	Yeom et al.	N/A	N/A
7803722	12/2009	Liang	N/A	N/A
D625977	12/2009	Watson et al.	N/A	N/A

7806587	12/2009	Kobayashi	N/A	N/A
7807566	12/2009	Tsuji et al.	N/A	N/A
7807578	12/2009	Bencher et al.	N/A	N/A
7816278	12/2009	Reed et al.	N/A	N/A
7824492	12/2009	Tois et al.	N/A	N/A
7825040	12/2009	Fukazawa et al.	N/A	N/A
7829460	12/2009	Streck et al.	N/A	N/A
7833348	12/2009	Wada et al.	N/A	N/A
7833353	12/2009	Furukawahara et al.	N/A	N/A
7838084	12/2009	Derderian et al.	N/A	N/A
7842518	12/2009	Miyajima	N/A	N/A
7842622	12/2009	Lee et al.	N/A	N/A
D629874	12/2009	Hermans	N/A	N/A
7850449	12/2009	Yang et al.	N/A	N/A
7851019	12/2009	Tuominen et al.	N/A	N/A
7851232	12/2009	van Schravendijk et al.	N/A	N/A
7858519	12/2009	Liu et al.	N/A	N/A
7858533	12/2009	Liu et al.	N/A	N/A
7858898	12/2009	Bailey et al.	N/A	N/A
7865070	12/2010	Nakamura	N/A	N/A
7871198	12/2010	Rempe et al.	N/A	N/A
7874726	12/2010	Jacobs et al.	N/A	N/A
7884918	12/2010	Hattori	N/A	N/A
7888233	12/2010	Gauri	N/A	N/A
7894474	12/2010	Bell	N/A	N/A
D633452	12/2010	Namiki et al.	N/A	N/A
D634329	12/2010	Wastrom	N/A	N/A
D634719	12/2010	Yasuda et al.	N/A	N/A
7897215	12/2010	Fair et al.	N/A	N/A
7897217	12/2010	Faguet	N/A	N/A
7902009	12/2010	Simonelli et al.	N/A	N/A
7902582	12/2010	Forbes et al.	N/A	N/A
7906174	12/2010	Wu et al.	N/A	N/A
7910288	12/2010	Abatchev et al.	N/A	N/A
7910452	12/2010	Roh et al.	N/A	N/A
7910494	12/2010	Dip et al.	N/A	N/A
7915139	12/2010	Lang	N/A	N/A
7915667	12/2010	Knoefler et al.	N/A	N/A
7919142	12/2010	Yeom et al.	N/A	N/A
7919416	12/2010	Lee et al.	N/A	N/A
7923382	12/2010	Huotari et al.	N/A	N/A
7925378	12/2010	Gilchrist et al.	N/A	N/A
7927395	12/2010	Szepessy et al.	N/A	N/A
7935940	12/2010	Smargiassi	N/A	N/A
7939447	12/2010	Bauer et al.	N/A	N/A
7942969	12/2010	Riker et al.	N/A	N/A
7946762	12/2010	Yednak	N/A	N/A
7951262	12/2010	Koshiishi et al.	N/A	N/A
7955516	12/2010	Chandrachood	N/A	N/A
7955646	12/2010	Cruse et al.	N/A	N/A

7955650	12/2010	Tsuji	N/A	N/A
7957708	12/2010	Karschnia et al.	N/A	N/A
7963736	12/2010	Takizawa et al.	N/A	N/A
7967913	12/2010	Hua et al.	N/A	N/A
7972980	12/2010	Lee et al.	N/A	N/A
7977256	12/2010	Liu et al.	N/A	N/A
7981751	12/2010	Zhu et al.	N/A	N/A
D643055	12/2010	Takahashi	N/A	N/A
7989365	12/2010	Park et al.	N/A	N/A
7989736	12/2010	Park et al.	N/A	N/A
7992318	12/2010	Kawaji	N/A	N/A
7993457	12/2010	Krotov et al.	N/A	N/A
7994070	12/2010	Dip et al.	N/A	N/A
7994721	12/2010	Espiau et al.	N/A	N/A
7997795	12/2010	Schwagerman et al.	N/A	N/A
7998875	12/2010	DeYoung	N/A	N/A
8003174	12/2010	Fukazawa	N/A	N/A
8003919	12/2010	Goto et al.	N/A	N/A
8004198	12/2010	Bakre et al.	N/A	N/A
8020315	12/2010	Nishimura	N/A	N/A
8030129	12/2010	Jeong	N/A	N/A
8033771	12/2010	Gage et al.	N/A	N/A
8038835	12/2010	Hayashi et al.	N/A	N/A
8041197	12/2010	Kasai et al.	N/A	N/A
8041450	12/2010	Takizawa et al.	N/A	N/A
8043972	12/2010	Liu et al.	N/A	N/A
8046193	12/2010	Yetter et al.	N/A	N/A
8047711	12/2010	Ploechinger	N/A	N/A
8048783	12/2010	Chung et al.	N/A	N/A
8051799	12/2010	Itagaki et al.	N/A	N/A
8052794	12/2010	Sumakeris et al.	N/A	N/A
8055378	12/2010	Numakura	N/A	N/A
8060252	12/2010	Gage et al.	N/A	N/A
RE43023	12/2010	Nakashima et al.	N/A	N/A
D649986	12/2010	Fujikata et al.	N/A	N/A
D651291	12/2010	Liebson et al.	N/A	N/A
8071451	12/2010	Berry	N/A	N/A
8071452	12/2010	Raisanen	N/A	N/A
8072578	12/2010	Yasuda et al.	N/A	N/A
8076230	12/2010	Wei	N/A	N/A
8076237	12/2010	Uzoh	N/A	N/A
8076250	12/2010	Rajagopalan	N/A	N/A
8076251	12/2010	Akae et al.	N/A	N/A
8078310	12/2010	Nishimoto et al.	N/A	N/A
8082946	12/2010	Laverdiere et al.	N/A	N/A
8083853	12/2010	Choi et al.	N/A	N/A
8084104	12/2010	Shinriki et al.	N/A	N/A
8084372	12/2010	You et al.	N/A	N/A
D652896	12/2011	Gether	N/A	N/A
8092604	12/2011	Tomiyasu et al.	N/A	N/A

8092606	12/2011	Park et al.	N/A	N/A
8100583	12/2011	Aggarwal	N/A	N/A
D653734	12/2011	Sisk	N/A	N/A
D654882	12/2011	Honma et al.	N/A	N/A
D654884	12/2011	Honma	N/A	N/A
D655055	12/2011	Toll	N/A	N/A
8110099	12/2011	Hersey et al.	N/A	N/A
8114734	12/2011	Yang et al.	N/A	N/A
8119466	12/2011	Avouris	N/A	N/A
8119527	12/2011	Chadrashekar et al.	N/A	N/A
D655260	12/2011	Honma et al.	N/A	N/A
D655261	12/2011	Honma et al.	N/A	N/A
D655599	12/2011	Durham	N/A	N/A
8128333	12/2011	Aburatani	N/A	N/A
8129290	12/2011	Balseanu et al.	N/A	N/A
8137462	12/2011	Fondurulia et al.	N/A	N/A
8137465	12/2011	Shrinivasan et al.	N/A	N/A
8138104	12/2011	Balseanu et al.	N/A	N/A
8138676	12/2011	Mills	N/A	N/A
8142862	12/2011	Lee et al.	N/A	N/A
8143174	12/2011	Xia et al.	N/A	N/A
8147242	12/2011	Shibagaki et al.	N/A	N/A
8152922	12/2011	Schmidt et al.	N/A	N/A
8158512	12/2011	Ji et al.	N/A	N/A
8172947	12/2011	Shibata et al.	N/A	N/A
8173554	12/2011	Lee et al.	N/A	N/A
8174400	12/2011	Park et al.	N/A	N/A
8178436	12/2011	King et al.	N/A	N/A
8187679	12/2011	Dickey et al.	N/A	N/A
8187951	12/2011	Wang	N/A	N/A
8192901	12/2011	Kageyama	N/A	N/A
8196234	12/2011	Glunk	N/A	N/A
8197915	12/2011	Oka et al.	N/A	N/A
8198168	12/2011	Tanioku	N/A	N/A
8202575	12/2011	Monsma et al.	N/A	N/A
8206506	12/2011	Kadkhodayan et al.	N/A	N/A
8216380	12/2011	White et al.	N/A	N/A
8227032	12/2011	Dussarrat et al.	N/A	N/A
8231799	12/2011	Bera et al.	N/A	N/A
D665055	12/2011	Yanagisawa et al.	N/A	N/A
8241991	12/2011	Hsieh et al.	N/A	N/A
8241992	12/2011	Clevenger et al.	N/A	N/A
8242028	12/2011	van Schravendijk et al.	N/A	N/A
8242031	12/2011	Mallick et al.	N/A	N/A
8246900	12/2011	Kasai et al.	N/A	N/A
8252114	12/2011	Vukovic	N/A	N/A
8252659	12/2011	Huyghabaert et al.	N/A	N/A
8252691	12/2011	Beynet et al.	N/A	N/A
8253204	12/2011	Lee et al.	N/A	N/A
8267633	12/2011	Obikane	N/A	N/A

8272516	12/2011	Salvador	N/A	N/A
8278176	12/2011	Bauer et al.	N/A	N/A
8278224	12/2011	Mui et al.	N/A	N/A
8282769	12/2011	Iizuka	N/A	N/A
8282847	12/2011	Romano	N/A	N/A
8282992	12/2011	Myo et al.	N/A	N/A
8287648	12/2011	Reed et al.	N/A	N/A
8291857	12/2011	Lam et al.	N/A	N/A
8293016	12/2011	Bahng et al.	N/A	N/A
8293642	12/2011	Kim	N/A	N/A
8298336	12/2011	Wang et al.	N/A	N/A
8298951	12/2011	Nakano	N/A	N/A
8307472	12/2011	Saxon et al.	N/A	N/A
8309173	12/2011	Tuominen et al.	N/A	N/A
8318327	12/2011	O'Donnell	N/A	N/A
8318584	12/2011	Li et al.	N/A	N/A
8323413	12/2011	Son	N/A	N/A
8324699	12/2011	Ichijo et al.	N/A	N/A
8328939	12/2011	Choi et al.	N/A	N/A
8329599	12/2011	Fukazawa et al.	N/A	N/A
8333839	12/2011	Oh	N/A	N/A
8334219	12/2011	Lee et al.	N/A	N/A
8338809	12/2011	Yang et al.	N/A	N/A
8349083	12/2012	Takasuka et al.	N/A	N/A
D676943	12/2012	Kluss et al.	N/A	N/A
8367528	12/2012	Bauer et al.	N/A	N/A
8372204	12/2012	Nakamura	N/A	N/A
8378464	12/2012	Kato et al.	N/A	N/A
8382370	12/2012	Aggarwal et al.	N/A	N/A
8382939	12/2012	Kutney et al.	N/A	N/A
8393091	12/2012	Kawamoto	N/A	N/A
8394466	12/2012	Hong et al.	N/A	N/A
8398773	12/2012	Jdira et al.	N/A	N/A
8402918	12/2012	Kadkhodayan et al.	N/A	N/A
8404044	12/2012	Arai	N/A	N/A
8404499	12/2012	Moffatt	N/A	N/A
8415258	12/2012	Akae	N/A	N/A
8415259	12/2012	Lee et al.	N/A	N/A
8415587	12/2012	Millman et al.	N/A	N/A
8419959	12/2012	Bettencourt et al.	N/A	N/A
8425682	12/2012	Wang et al.	N/A	N/A
8430620	12/2012	Blank et al.	N/A	N/A
8435894	12/2012	Chandrashekar et al.	N/A	N/A
8440259	12/2012	Chiang et al.	N/A	N/A
8443484	12/2012	Ozaki et al.	N/A	N/A
8444120	12/2012	Gregg et al.	N/A	N/A
8445075	12/2012	Xu et al.	N/A	N/A
8450191	12/2012	Wang	N/A	N/A
8454749	12/2012	Li	N/A	N/A
8465811	12/2012	Ueda	N/A	N/A

8465903	12/2012	Weidman et al.	N/A	N/A
8466411	12/2012	Arai	N/A	N/A
8470187	12/2012	Ha	N/A	N/A
8470718	12/2012	Lee	N/A	N/A
8484846	12/2012	Dhindsa	N/A	N/A
8492170	12/2012	Xie et al.	N/A	N/A
8496377	12/2012	Harr et al.	N/A	N/A
8496756	12/2012	Cruse et al.	N/A	N/A
8497213	12/2012	Yasui et al.	N/A	N/A
8501599	12/2012	Ueno et al.	N/A	N/A
8506162	12/2012	Schick et al.	N/A	N/A
8506713	12/2012	Takagi	N/A	N/A
8507720	12/2012	Shay	N/A	N/A
8524612	12/2012	Li et al.	N/A	N/A
8529701	12/2012	Morita	N/A	N/A
8535767	12/2012	Kimura	N/A	N/A
8536068	12/2012	Weidman et al.	N/A	N/A
D691974	12/2012	Osada et al.	N/A	N/A
8551892	12/2012	Nakano	N/A	N/A
8557712	12/2012	Antonelli et al.	N/A	N/A
8562272	12/2012	Lenz	N/A	N/A
8563443	12/2012	Fukazawa	N/A	N/A
8569184	12/2012	Oka	N/A	N/A
D693200	12/2012	Saunders	N/A	N/A
D693782	12/2012	Mori et al.	N/A	N/A
8573152	12/2012	de La Llera et al.	N/A	N/A
8573154	12/2012	Yorozuya	N/A	N/A
8586484	12/2012	Matsuyama et al.	N/A	N/A
8591659	12/2012	Fang et al.	N/A	N/A
8592005	12/2012	Ueda	N/A	N/A
D694790	12/2012	Matsumoto et al.	N/A	N/A
D695240	12/2012	Iida et al.	N/A	N/A
8608885	12/2012	Goto et al.	N/A	N/A
8614047	12/2012	Ayothi et al.	N/A	N/A
8616765	12/2012	Darabnia et al.	N/A	N/A
8617411	12/2012	Singh	N/A	N/A
D697038	12/2013	Matsumoto et al.	N/A	N/A
8623770	12/2013	Gao et al.	N/A	N/A
8633115	12/2013	Chang et al.	N/A	N/A
8637384	12/2013	Ando et al.	N/A	N/A
D698904	12/2013	Milligan et al.	N/A	N/A
8642488	12/2013	Liu et al.	N/A	N/A
8647439	12/2013	Sanchez et al.	N/A	N/A
8647722	12/2013	Kobayashi et al.	N/A	N/A
8647993	12/2013	Lavoie et al.	N/A	N/A
8651788	12/2013	Budde	N/A	N/A
8657942	12/2013	Shaw et al.	N/A	N/A
8664127	12/2013	Bhatia et al.	N/A	N/A
8664627	12/2013	Ishikawa et al.	N/A	N/A
8667654	12/2013	Gros-Jean	N/A	N/A

8668957	12/2013	Dussarrat et al.	N/A	N/A
8669185	12/2013	Onizawa	N/A	N/A
8679958	12/2013	Takamure et al.	N/A	N/A
D702188	12/2013	Jacobs	N/A	N/A
8683943	12/2013	Onodera et al.	N/A	N/A
8697198	12/2013	Lee	N/A	N/A
8703002	12/2013	Matsudo et al.	N/A	N/A
8709162	12/2013	Leung et al.	N/A	N/A
8710580	12/2013	Sakuma et al.	N/A	N/A
8711338	12/2013	Liu et al.	N/A	N/A
D705745	12/2013	Kurs et al.	N/A	N/A
D705762	12/2013	Yu	N/A	N/A
8720965	12/2013	Hino et al.	N/A	N/A
8721791	12/2013	Choi et al.	N/A	N/A
8722510	12/2013	Watanabe et al.	N/A	N/A
8722546	12/2013	Fukazawa et al.	N/A	N/A
8726837	12/2013	Patalay et al.	N/A	N/A
8728832	12/2013	Raisanen et al.	N/A	N/A
8728956	12/2013	Lavoie et al.	N/A	N/A
8741062	12/2013	Lindfors et al.	N/A	N/A
8741065	12/2013	Odagiri et al.	N/A	N/A
8742668	12/2013	Nakano et al.	N/A	N/A
8758512	12/2013	Lee	N/A	N/A
8759223	12/2013	Sapre et al.	N/A	N/A
D709536	12/2013	Yoshimura et al.	N/A	N/A
D709537	12/2013	Kuwabara et al.	N/A	N/A
8764085	12/2013	Urabe	N/A	N/A
8771791	12/2013	Lee et al.	N/A	N/A
8771807	12/2013	Xiao et al.	N/A	N/A
8779502	12/2013	Sakuma et al.	N/A	N/A
8784676	12/2013	Guha et al.	N/A	N/A
8784950	12/2013	Fukazawa et al.	N/A	N/A
8784951	12/2013	Fukazawa et al.	N/A	N/A
8785215	12/2013	Kobayashi et al.	N/A	N/A
8785311	12/2013	Miyoshi	N/A	N/A
8790743	12/2013	Omari	N/A	N/A
8790749	12/2013	Omori et al.	N/A	N/A
8802201	12/2013	Raisanen et al.	N/A	N/A
8809170	12/2013	Bauer	N/A	N/A
D712358	12/2013	Allen et al.	N/A	N/A
D712359	12/2013	Allen et al.	N/A	N/A
8820809	12/2013	Ando et al.	N/A	N/A
8821640	12/2013	Cleary et al.	N/A	N/A
8821985	12/2013	Shao et al.	N/A	N/A
8823672	12/2013	Kim	N/A	N/A
8828886	12/2013	Samukawa et al.	N/A	N/A
8841182	12/2013	Chen et al.	N/A	N/A
8845806	12/2013	Aida et al.	N/A	N/A
8846502	12/2013	Haukka et al.	N/A	N/A
D715410	12/2013	Lohmann	N/A	N/A

8859368	12/2013	Deniz	N/A	N/A
8860955	12/2013	Rodnick et al.	N/A	N/A
8864202	12/2013	Schrameyer	N/A	N/A
8864375	12/2013	Abe et al.	N/A	N/A
D716742	12/2013	Jang et al.	N/A	N/A
8876974	12/2013	Wan	N/A	N/A
8877300	12/2013	Lee	N/A	N/A
8877655	12/2013	Shero et al.	N/A	N/A
8882923	12/2013	Saido et al.	N/A	N/A
8883270	12/2013	Shero et al.	N/A	N/A
8895108	12/2013	Lee	N/A	N/A
8895395	12/2013	Kerber et al.	N/A	N/A
8900935	12/2013	Guo et al.	N/A	N/A
8900999	12/2013	Wu et al.	N/A	N/A
8901016	12/2013	Jeongseok et al.	N/A	N/A
8911553	12/2013	Baluja et al.	N/A	N/A
8911826	12/2013	Adachi et al.	N/A	N/A
8912101	12/2013	Tsuji et al.	N/A	N/A
D720838	12/2014	Yamagishi et al.	N/A	N/A
8927906	12/2014	Tadokoro et al.	N/A	N/A
8932802	12/2014	Wu et al.	N/A	N/A
8933375	12/2014	Dunn et al.	N/A	N/A
8937800	12/2014	Lubomirsky et al.	N/A	N/A
8940646	12/2014	Chandrasekharan	N/A	N/A
D723153	12/2014	Borkholder	N/A	N/A
8945305	12/2014	Marsh	N/A	N/A
8945306	12/2014	Tsuda	N/A	N/A
8945339	12/2014	Kakimoto	N/A	N/A
8946830	12/2014	Jung et al.	N/A	N/A
8956971	12/2014	Huakka	N/A	N/A
8956983	12/2014	Swaminathan	N/A	N/A
D723330	12/2014	York	N/A	N/A
D724553	12/2014	Choi	N/A	N/A
D724701	12/2014	Yamagishi et al.	N/A	N/A
D725168	12/2014	Yamagishi	N/A	N/A
8967608	12/2014	Mitsumori et al.	N/A	N/A
8968989	12/2014	Ouattara et al.	N/A	N/A
8969934	12/2014	Cheng et al.	N/A	N/A
8974868	12/2014	Ishikawa et al.	N/A	N/A
8980006	12/2014	Huh et al.	N/A	N/A
8986456	12/2014	Fondurulia et al.	N/A	N/A
8991214	12/2014	Hoshino et al.	N/A	N/A
8991887	12/2014	Shin et al.	N/A	N/A
8993054	12/2014	Jung et al.	N/A	N/A
8993072	12/2014	Xiao et al.	N/A	N/A
8993457	12/2014	Ramkumar et al.	N/A	N/A
D726365	12/2014	Weigensberg	N/A	N/A
D726884	12/2014	Yamagishi et al.	N/A	N/A
8999102	12/2014	Miyoshi et al.	N/A	N/A
9004744	12/2014	Kemp	N/A	N/A

9005539	12/2014	Halpin et al.	N/A	N/A
9017481	12/2014	Pettinger et al.	N/A	N/A
9017933	12/2014	Liu et al.	N/A	N/A
9018093	12/2014	Tsuji et al.	N/A	N/A
9018111	12/2014	Milligan et al.	N/A	N/A
9018567	12/2014	de Ridder et al.	N/A	N/A
9021985	12/2014	Alokozai et al.	N/A	N/A
9023737	12/2014	Beynet et al.	N/A	N/A
9023738	12/2014	Kato et al.	N/A	N/A
9029244	12/2014	Won et al.	N/A	N/A
9029253	12/2014	Milligan et al.	N/A	N/A
9029272	12/2014	Nakano	N/A	N/A
D732145	12/2014	Yamagishi	N/A	N/A
D732644	12/2014	Yamagishi et al.	N/A	N/A
D733257	12/2014	Schoenherr et al.	N/A	N/A
D733261	12/2014	Yamagishi et al.	N/A	N/A
D733262	12/2014	Kang	N/A	N/A
9057388	12/2014	Comeau et al.	N/A	N/A
9064815	12/2014	Zhang et al.	N/A	N/A
D733843	12/2014	Yamagishi	N/A	N/A
D734377	12/2014	Hirakida	N/A	N/A
9076635	12/2014	Gross et al.	N/A	N/A
9076726	12/2014	Kauerauf et al.	N/A	N/A
D735836	12/2014	Yamagishi et al.	N/A	N/A
D736348	12/2014	Tan	N/A	N/A
9095869	12/2014	Kilpi et al.	N/A	N/A
9096931	12/2014	Yednak et al.	N/A	N/A
9099423	12/2014	Weeks et al.	N/A	N/A
9099505	12/2014	Kusakabe et al.	N/A	N/A
9111972	12/2014	Takeshita et al.	N/A	N/A
9117657	12/2014	Nakano et al.	N/A	N/A
9117866	12/2014	Marquardt et al.	N/A	N/A
D739222	12/2014	Chadbourne	N/A	N/A
9123510	12/2014	Nakano et al.	N/A	N/A
9123577	12/2014	Fujimoto et al.	N/A	N/A
9127358	12/2014	Inoue et al.	N/A	N/A
9127362	12/2014	Scheible et al.	N/A	N/A
9129897	12/2014	Pore et al.	N/A	N/A
9136108	12/2014	Matsushita et al.	N/A	N/A
9136180	12/2014	Machkaoutsan	N/A	N/A
9142393	12/2014	Okabe et al.	N/A	N/A
9142437	12/2014	Fosnight et al.	N/A	N/A
9142764	12/2014	Wang	N/A	N/A
9153441	12/2014	Takamure et al.	N/A	N/A
9163310	12/2014	Lee	N/A	N/A
9166012	12/2014	Sim et al.	N/A	N/A
9169975	12/2014	Sarin et al.	N/A	N/A
9171714	12/2014	Mori	N/A	N/A
9171715	12/2014	Matero	N/A	N/A
9171716	12/2014	Fukuda	N/A	N/A

D742202	12/2014	Cyphers et al.	N/A	N/A
D743357	12/2014	Vyne	N/A	N/A
D743513	12/2014	Yamagishi	N/A	N/A
9174178	12/2014	Berger et al.	N/A	N/A
9175394	12/2014	Yudovsky et al.	N/A	N/A
9177784	12/2014	Raisanen et al.	N/A	N/A
9184047	12/2014	Liu et al.	N/A	N/A
9184054	12/2014	Huang et al.	N/A	N/A
9190263	12/2014	Ishikawa et al.	N/A	N/A
9190264	12/2014	Yuasa et al.	N/A	N/A
9196483	12/2014	Lee et al.	N/A	N/A
D745641	12/2014	Blum	N/A	N/A
9202727	12/2014	Dunn et al.	N/A	N/A
9214333	12/2014	Sims et al.	N/A	N/A
9214340	12/2014	Kurita et al.	N/A	N/A
9219006	12/2014	Chatterjee	N/A	N/A
9223203	12/2014	Farm et al.	N/A	N/A
9228259	12/2015	Haukka et al.	N/A	N/A
9240412	12/2015	Xie et al.	N/A	N/A
9245742	12/2015	Haukka	N/A	N/A
9252024	12/2015	Lam et al.	N/A	N/A
9252238	12/2015	Trevino et al.	N/A	N/A
9257274	12/2015	Kang et al.	N/A	N/A
9263298	12/2015	Matsumoto et al.	N/A	N/A
9267204	12/2015	Honma	N/A	N/A
9267850	12/2015	Aggarwal	N/A	N/A
D751176	12/2015	Schoenherr et al.	N/A	N/A
9275834	12/2015	Park et al.	N/A	N/A
9281223	12/2015	Hara	N/A	N/A
9281277	12/2015	Baek et al.	N/A	N/A
9284642	12/2015	Nakano	N/A	N/A
9287273	12/2015	Ragnarsson et al.	N/A	N/A
9297705	12/2015	Aggarwal	N/A	N/A
9299557	12/2015	Tolle et al.	N/A	N/A
9299595	12/2015	Dunn et al.	N/A	N/A
D753269	12/2015	Yamagishi et al.	N/A	N/A
D753629	12/2015	Plattard	N/A	N/A
9305836	12/2015	Gates et al.	N/A	N/A
9309598	12/2015	Wang et al.	N/A	N/A
9309978	12/2015	Hatch et al.	N/A	N/A
9310684	12/2015	Meyers et al.	N/A	N/A
9312155	12/2015	Mori	N/A	N/A
9315897	12/2015	Byun	N/A	N/A
9324811	12/2015	Weeks	N/A	N/A
9324846	12/2015	Camillo	N/A	N/A
D756929	12/2015	Harck et al.	N/A	N/A
9331200	12/2015	Wang et al.	N/A	N/A
9337031	12/2015	Kim et al.	N/A	N/A
9337054	12/2015	Hunks et al.	N/A	N/A
9337057	12/2015	Park et al.	N/A	N/A

9341296	12/2015	Yednak	N/A	N/A
9343297	12/2015	Fukazawa et al.	N/A	N/A
9343308	12/2015	Isii	N/A	N/A
9343343	12/2015	Mori	N/A	N/A
9343350	12/2015	Arai	N/A	N/A
9349620	12/2015	Kamata et al.	N/A	N/A
9353440	12/2015	Ge et al.	N/A	N/A
9353441	12/2015	Chung	N/A	N/A
9355876	12/2015	Reuter et al.	N/A	N/A
9355882	12/2015	Wu et al.	N/A	N/A
D759137	12/2015	Hassan	N/A	N/A
D759193	12/2015	Gutierrez et al.	N/A	N/A
9362107	12/2015	Thadani et al.	N/A	N/A
9362137	12/2015	Kang et al.	N/A	N/A
9362180	12/2015	Lee et al.	N/A	N/A
9365924	12/2015	Nonaka	N/A	N/A
9368352	12/2015	Takamure et al.	N/A	N/A
9370863	12/2015	Tsuji et al.	N/A	N/A
9378969	12/2015	Hsu et al.	N/A	N/A
D761325	12/2015	Abed	N/A	N/A
9384987	12/2015	Jung et al.	N/A	N/A
9390909	12/2015	Pasquale et al.	N/A	N/A
9394608	12/2015	Shero et al.	N/A	N/A
9396934	12/2015	Tolle	N/A	N/A
9396956	12/2015	Fukazawa	N/A	N/A
9399228	12/2015	Breiling et al.	N/A	N/A
D764196	12/2015	Handler et al.	N/A	N/A
9404587	12/2015	Shugrue	N/A	N/A
9410247	12/2015	Lee et al.	N/A	N/A
9412564	12/2015	Milligan	N/A	N/A
9412581	12/2015	Thadani et al.	N/A	N/A
9412582	12/2015	Sasaki et al.	N/A	N/A
9418885	12/2015	Sung et al.	N/A	N/A
9425078	12/2015	Tang et al.	N/A	N/A
9428833	12/2015	Duvall et al.	N/A	N/A
9443725	12/2015	Liu et al.	N/A	N/A
9447498	12/2015	Shiba et al.	N/A	N/A
9449793	12/2015	Shaji et al.	N/A	N/A
9449795	12/2015	Sabri et al.	N/A	N/A
9449843	12/2015	Korolik et al.	N/A	N/A
9449987	12/2015	Miyata et al.	N/A	N/A
9455138	12/2015	Fukazawa	N/A	N/A
9455177	12/2015	Park et al.	N/A	N/A
9460954	12/2015	De Jong et al.	N/A	N/A
9464352	12/2015	Nakano et al.	N/A	N/A
9472410	12/2015	Sadjadi et al.	N/A	N/A
9472432	12/2015	Blank	N/A	N/A
9474163	12/2015	Tolle et al.	N/A	N/A
9478414	12/2015	Kobayashi et al.	N/A	N/A
9478415	12/2015	Kimura	N/A	N/A

D770993	12/2015	Yoshida et al.	N/A	N/A
9484191	12/2015	Winkler	N/A	N/A
9496225	12/2015	Adusumilli et al.	N/A	N/A
9514927	12/2015	Tolle et al.	N/A	N/A
9514932	12/2015	Mallick et al.	N/A	N/A
9520289	12/2015	Park et al.	N/A	N/A
9523148	12/2015	Pore et al.	N/A	N/A
D777546	12/2016	Ishii et al.	N/A	N/A
9543180	12/2016	Kamiya	N/A	N/A
9556516	12/2016	Takamure	N/A	N/A
9558931	12/2016	Tang	N/A	N/A
9564312	12/2016	Henri et al.	N/A	N/A
9564314	12/2016	Takamure et al.	N/A	N/A
9570302	12/2016	Chang et al.	N/A	N/A
9574268	12/2016	Dunn et al.	N/A	N/A
9576952	12/2016	Joshi et al.	N/A	N/A
9583333	12/2016	Chatterjee	N/A	N/A
9583345	12/2016	Chen et al.	N/A	N/A
D782419	12/2016	Willette	N/A	N/A
9589770	12/2016	Winkler	N/A	N/A
9605342	12/2016	Alokozai et al.	N/A	N/A
9605343	12/2016	Winkler	N/A	N/A
9605736	12/2016	Foshage et al.	N/A	N/A
9607837	12/2016	Namba	N/A	N/A
D783351	12/2016	Fujino et al.	N/A	N/A
D784276	12/2016	Tiner et al.	N/A	N/A
9613801	12/2016	Carcasi et al.	N/A	N/A
9618846	12/2016	Shamma et al.	N/A	N/A
9627221	12/2016	Zaitzu et al.	N/A	N/A
D785766	12/2016	Sato	N/A	N/A
D787458	12/2016	Kim et al.	N/A	N/A
9640416	12/2016	Arai	N/A	N/A
9640448	12/2016	Ikegawa et al.	N/A	N/A
9640542	12/2016	Lee et al.	N/A	N/A
9644266	12/2016	Nasu et al.	N/A	N/A
9647114	12/2016	Margetis	N/A	N/A
9653267	12/2016	Carducci et al.	N/A	N/A
9657845	12/2016	Shugrue	N/A	N/A
9659799	12/2016	Lawson	N/A	N/A
9663857	12/2016	Nakano et al.	N/A	N/A
9666528	12/2016	Bergendahl et al.	N/A	N/A
D789888	12/2016	Jang et al.	N/A	N/A
D790041	12/2016	Jang et al.	N/A	N/A
9680268	12/2016	Finona	N/A	N/A
9684234	12/2016	Darling et al.	N/A	N/A
9685320	12/2016	Kang et al.	N/A	N/A
9691668	12/2016	Chang et al.	N/A	N/A
9691771	12/2016	Lansalot-Matras	N/A	N/A
9698031	12/2016	Kobayashi et al.	N/A	N/A
9708707	12/2016	Ditizio et al.	N/A	N/A

9708708	12/2016	Isobe et al.	N/A	N/A
9711345	12/2016	Shiba et al.	N/A	N/A
D793352	12/2016	Hill	N/A	N/A
D793526	12/2016	Behdjat	N/A	N/A
D793572	12/2016	Kozuka et al.	N/A	N/A
D793976	12/2016	Fukushima et al.	N/A	N/A
D794753	12/2016	Miller	N/A	N/A
D795208	12/2016	Sasaki et al.	N/A	N/A
9735024	12/2016	Zaitsu	N/A	N/A
9741546	12/2016	Carducci et al.	N/A	N/A
9741559	12/2016	Shimura et al.	N/A	N/A
9745658	12/2016	Kang et al.	N/A	N/A
9748104	12/2016	Sasaki et al.	N/A	N/A
9748145	12/2016	Kannan et al.	N/A	N/A
D796458	12/2016	Jang et al.	N/A	N/A
D796670	12/2016	Dolk et al.	N/A	N/A
D797067	12/2016	Zhang et al.	N/A	N/A
D798248	12/2016	Hanson et al.	N/A	N/A
9754779	12/2016	Ishikawa	N/A	N/A
9754818	12/2016	Shiu et al.	N/A	N/A
9759489	12/2016	Kaneko	N/A	N/A
9765432	12/2016	Ge et al.	N/A	N/A
9773818	12/2016	Kimura	N/A	N/A
D800782	12/2016	Bever et al.	N/A	N/A
9778561	12/2016	Marks et al.	N/A	N/A
9780225	12/2016	Venkatasubramanian et al.	N/A	N/A
9786491	12/2016	Suzuki et al.	N/A	N/A
9786570	12/2016	Kang et al.	N/A	N/A
9790595	12/2016	Jung et al.	N/A	N/A
9793115	12/2016	Tolle	N/A	N/A
9793135	12/2016	Zaitsu et al.	N/A	N/A
9793148	12/2016	Yamagishi et al.	N/A	N/A
9798308	12/2016	Mimura	N/A	N/A
9799736	12/2016	Ebrish et al.	N/A	N/A
9803926	12/2016	Kikuchi et al.	N/A	N/A
D801942	12/2016	Riker et al.	N/A	N/A
D802472	12/2016	Sasaki et al.	N/A	N/A
D802546	12/2016	Jang et al.	N/A	N/A
D803802	12/2016	Sasaki et al.	N/A	N/A
9808246	12/2016	Shelton et al.	N/A	N/A
9812319	12/2016	Fukazawa et al.	N/A	N/A
9812320	12/2016	Pore et al.	N/A	N/A
9812372	12/2016	Choi et al.	N/A	N/A
9820289	12/2016	Pawar et al.	N/A	N/A
9824881	12/2016	Niskanen et al.	N/A	N/A
9824884	12/2016	Sims et al.	N/A	N/A
9824893	12/2016	Smith et al.	N/A	N/A
9837355	12/2016	Briggs et al.	N/A	N/A
9842835	12/2016	Cheng et al.	N/A	N/A

9847221	12/2016	McLaughlin et al.	N/A	N/A
9847247	12/2016	Huang et al.	N/A	N/A
9850573	12/2016	Sun	N/A	N/A
D807494	12/2017	Kim et al.	N/A	N/A
D808254	12/2017	Deleu	N/A	N/A
9859151	12/2017	Zhu	N/A	N/A
9865455	12/2017	Sims et al.	N/A	N/A
9865456	12/2017	Pandey et al.	N/A	N/A
9865815	12/2017	Hausmann	N/A	N/A
9868131	12/2017	Kilpi et al.	N/A	N/A
9870964	12/2017	Yoshino et al.	N/A	N/A
9875891	12/2017	Henri et al.	N/A	N/A
9875893	12/2017	Takamure et al.	N/A	N/A
9881788	12/2017	Kim et al.	N/A	N/A
D810705	12/2017	Krishnan et al.	N/A	N/A
9887082	12/2017	Pore et al.	N/A	N/A
9890456	12/2017	Tolle et al.	N/A	N/A
9891521	12/2017	Kang et al.	N/A	N/A
9892908	12/2017	Pettinger et al.	N/A	N/A
9892913	12/2017	Margetis et al.	N/A	N/A
9895715	12/2017	Haukka et al.	N/A	N/A
9896761	12/2017	Hara et al.	N/A	N/A
9899291	12/2017	Kato	N/A	N/A
9899405	12/2017	Kim	N/A	N/A
9905420	12/2017	Margetis et al.	N/A	N/A
9905492	12/2017	Tang et al.	N/A	N/A
9909214	12/2017	Suemori	N/A	N/A
9909492	12/2017	Jeswine	N/A	N/A
9911595	12/2017	Smith et al.	N/A	N/A
9911676	12/2017	Tang	N/A	N/A
9916980	12/2017	Knaepen	N/A	N/A
9920451	12/2017	Sivaramakrishnan et al.	N/A	N/A
9922824	12/2017	Okada	N/A	N/A
9929005	12/2017	Shimamoto et al.	N/A	N/A
9929011	12/2017	Hawryluk et al.	N/A	N/A
9929055	12/2017	Dube et al.	N/A	N/A
9951421	12/2017	Lind	N/A	N/A
9960033	12/2017	Nozawa	N/A	N/A
9960072	12/2017	Coomer	N/A	N/A
9966299	12/2017	Tang et al.	N/A	N/A
9970112	12/2017	Koshi et al.	N/A	N/A
9984869	12/2017	Blanquart	N/A	N/A
D819580	12/2017	Krishnan et al.	N/A	N/A
9987747	12/2017	Hwang et al.	N/A	N/A
9991138	12/2017	Lin et al.	N/A	N/A
9996004	12/2017	Smith et al.	N/A	N/A
9997357	12/2017	Arghavani et al.	N/A	N/A
9997373	12/2017	Hudson	N/A	N/A
10014212	12/2017	Chen et al.	N/A	N/A
10017856	12/2017	Arnepalli et al.	N/A	N/A

10018920	12/2017	Chang et al.	N/A	N/A
10023960	12/2017	Alokozai	N/A	N/A
10032628	12/2017	Xie et al.	N/A	N/A
10032792	12/2017	Kim et al.	N/A	N/A
D825505	12/2017	Hanson et al.	N/A	N/A
D825614	12/2017	Bever et al.	N/A	N/A
10043661	12/2017	Kato et al.	N/A	N/A
10047435	12/2017	Haukka et al.	N/A	N/A
10053774	12/2017	Tolle et al.	N/A	N/A
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10087522	12/2017	Raisanen et al.	N/A	N/A
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10090316	12/2017	Ootsuka	N/A	N/A
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10177025	12/2018	Pore	N/A	N/A
10179947	12/2018	Fukazawa	N/A	N/A
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10190213	12/2018	Zhu et al.	N/A	N/A
10190214	12/2018	Shon et al.	N/A	N/A
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10229833	12/2018	Raisanen et al.	N/A	N/A
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10249524	12/2018	den Hartog Besselink et al.	N/A	N/A
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10269558	12/2018	Blanquart et al.	N/A	N/A
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10319588	12/2018	Mattinen et al.	N/A	N/A
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10332963	12/2018	Xie	N/A	N/A
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10340135	12/2018	Blanquart	N/A	N/A
10343920	12/2018	Haukka	N/A	N/A
10347547	12/2018	Varadarajan et al.	N/A	N/A
10354873	12/2018	Ko et al.	N/A	N/A
10361201	12/2018	Xie et al.	N/A	N/A
10361366	12/2018	Hakamata et al.	N/A	N/A
10367080	12/2018	Tang et al.	N/A	N/A
10388513	12/2018	Blanquart	N/A	N/A
10395917	12/2018	Niskanen et al.	N/A	N/A
10395919	12/2018	Masaru et al.	N/A	N/A
10395963	12/2018	Cooke	N/A	N/A
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10400335	12/2018	Ge et al.	N/A	N/A
10410943	12/2018	Jiang et al.	N/A	N/A
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10428419	12/2018	Huotari et al.	N/A	N/A
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10529554	12/2019	Ishikawa et al.	N/A	N/A
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10590531	12/2019	Shirako et al.	N/A	N/A
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10622196	12/2019	Nagayama et al.	N/A	N/A
10622236	12/2019	Kuo et al.	N/A	N/A

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10662525	12/2019	Jang et al.	N/A	N/A
10704143	12/2019	Hisamitsu et al.	N/A	N/A
10714335	12/2019	Kim et al.	N/A	N/A
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10770336	12/2019	Hill et al.	N/A	N/A
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2009/0104594	12/2008	Webb	N/A	N/A
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2009/0107404	12/2008	Ogliari et al.	N/A	N/A
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2009/0112458	12/2008	Nakai	N/A	N/A
2009/0115064	12/2008	Sandhu et al.	N/A	N/A
2009/0116936	12/2008	Marubayashi et al.	N/A	N/A
2009/0117717	12/2008	Tomasini et al.	N/A	N/A
2009/0117723	12/2008	Kim et al.	N/A	N/A
2009/0117746	12/2008	Masuda	N/A	N/A

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2009/0122293	12/2008	Shibazaki	N/A	N/A
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2009/0130331	12/2008	Asai	N/A	N/A
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2009/0142905	12/2008	Yamazaki	N/A	N/A
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2009/0165715	12/2008	Oh	N/A	N/A
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2009/0166616	12/2008	Uchiyama	N/A	N/A
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2011/0117728	12/2010	Su et al.	N/A	N/A
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2011/0117749	12/2010	Sheu	N/A	N/A
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2011/0159673	12/2010	Hanawa et al.	N/A	N/A
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2011/0168330	12/2010	Sakaue et al.	N/A	N/A
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2013/0075746	12/2012	Mallikarjunaswamy et al.	N/A	N/A
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2013/0078376	12/2012	Higashino et al.	N/A	N/A
2013/0078392	12/2012	Xiao et al.	N/A	N/A
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2013/0082274	12/2012	Yang	N/A	N/A
2013/0084156	12/2012	Shimamoto	N/A	N/A
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2013/0084711	12/2012	Liang et al.	N/A	N/A
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2013/0085618	12/2012	Ding	N/A	N/A
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2013/0089716	12/2012	Krishnamurthy et al.	N/A	N/A
2013/0089988	12/2012	Wang et al.	N/A	N/A
2013/0092085	12/2012	Lee	N/A	N/A
2013/0093048	12/2012	Chang et al.	N/A	N/A
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2013/0113085	12/2012	Michaelson et al.	N/A	N/A
2013/0115383	12/2012	Lu et al.	N/A	N/A
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2013/0115768	12/2012	Pore et al.	N/A	N/A
2013/0118405	12/2012	Ho et al.	N/A	N/A
2013/0118895	12/2012	Roozeboom et al.	N/A	N/A
2013/0119018	12/2012	Kanarik et al.	N/A	N/A
2013/0122712	12/2012	Kim et al.	N/A	N/A
2013/0122722	12/2012	Cissell et al.	N/A	N/A
2013/0126515	12/2012	Shero et al.	N/A	N/A
2013/0129577	12/2012	Halpin et al.	N/A	N/A
2013/0130490	12/2012	Lee et al.	N/A	N/A
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2013/0134148	12/2012	Tachikawa	N/A	N/A
2013/0137279	12/2012	Yamamoto et al.	N/A	N/A
2013/0143401	12/2012	Yu et al.	N/A	N/A
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2013/0228225	12/2012	Leeser	N/A	N/A
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2013/0285155	12/2012	Glass	N/A	N/A
2013/0287526	12/2012	Bluck et al.	N/A	N/A
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2015/0041431	12/2014	Zafiropoulo et al.	N/A	N/A
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2015/0099342	12/2014	Tsai	N/A	N/A
2015/0099374	12/2014	Kakimoto et al.	N/A	N/A
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2015/0110147	12/2014	Mizuno	N/A	N/A
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2015/0111374	12/2014	Bao	N/A	N/A
2015/0111395	12/2014	Hashimoto et al.	N/A	N/A
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2017/0036935	12/2016	Peterson	N/A	N/A
2017/0037513	12/2016	Haukka	N/A	N/A
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2017/0092847	12/2016	Kim et al.	N/A	N/A
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2017/0114464	12/2016	Iriuda et al.	N/A	N/A
2017/0114465	12/2016	Kalutarage et al.	N/A	N/A
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2017/0154770	12/2016	Margetis et al.	N/A	N/A
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2017/0260649	12/2016	Coomer	N/A	N/A
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2018/0105930	12/2017	Kang et al.	N/A	N/A
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USPTO; Notice of Allowance dated Dec. 10, 2018 in U.S. Appl. No. 15/863,340. cited by applicant
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USPTO; Advisory Action dated Nov. 5, 2019 in U.S. Appl. No. 15/879,209. cited by applicant
USPTO; Notice of Allowance dated Jan. 16, 2020 in U.S. Appl. No. 15/879,209. cited by applicant
USPTO; Non-Final Office Action dated Apr. 17, 2019 in U.S. Appl. No. 15/886,225. cited by applicant
USPTO; Notice of Allowance dated Sep. 23, 2019 in U.S. Appl. No. 15/886,225. cited by applicant

USPTO; Non-Final Office Action dated Nov. 15, 2018 in U.S. Appl. No. 15/890,037. cited by applicant

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USPTO; Notice of Allowance dated Feb. 8, 2019 in U.S. Appl. No. 15/892,756. cited by applicant

USPTO; Non-Final Office Action dated Apr. 24, 2019 in U.S. Appl. No. 15/896,986. cited by applicant

USPTO; Final Office Action dated Nov. 19, 2019 in U.S. Appl. No. 15/896,986. cited by applicant

USPTO; Advisory Action dated Feb. 7, 2020 in U.S. Appl. No. 15/896,986. cited by applicant

USPTO; Non-Final Office Action dated Mar. 24, 2020 in U.S. Appl. No. 15/896,986. cited by applicant

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USPTO; Notice of Allowance dated Aug. 26, 2020 in U.S. Appl. No. 15/896,986. cited by applicant

USPTO; Non-Final Office Action dated Sep. 19, 2019 in U.S. Appl. No. 15/897,578. cited by applicant

USPTO; Notice of Allowance dated Mar. 27, 2020 in U.S. Appl. No. 15/897,578. cited by applicant

USPTO; Non-Final Office Action dated May 30, 2019 in U.S. Appl. No. 15/900,425. cited by applicant

USPTO; Notice of Allowance dated Dec. 17, 2019 in U.S. Appl. No. 15/900,425. cited by applicant

USPTO; Non-Final Office Action dated Sep. 9, 2020 in U.S. Appl. No. 15/909,705. cited by applicant

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USPTO; Non-Final Office Action dated May 1, 2020 in U.S. Appl. No. 15/917,224. cited by applicant

USPTO; Final Office Action dated Sep. 24, 2020 in U.S. Appl. No. 15/917,224. cited by applicant

USPTO; Non-Final Office Action dated Feb. 8, 2019 in U.S. Appl. No. 15/917,262. cited by applicant

USPTO; Final Office Action dated Jun. 14, 2019 in U.S. Appl. No. 15/917,262. cited by applicant

USPTO; Notice of Allowance dated Aug. 30, 2019 in U.S. Appl. No. 15/917,262. cited by applicant

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USPTO; Advisory Action dated May 15, 2020 in U.S. Appl. No. 15/923,834. cited by applicant

USPTO; Non-Final Office Action dated Jul. 6, 2020 in U.S. Appl. No. 15/923,834. cited by applicant

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USPTO; Non-Final Office Action dated May 8, 2019 in U.S. Appl. No. 15/925,532. cited by applicant

USPTO; Final Office Action dated Nov. 27, 2019 in U.S. Appl. No. 15/925,532. cited by applicant

USPTO; Non-Final Office Action dated May 1, 2020 in U.S. Appl. No. 15/925,532. cited by applicant

USPTO; Final Office Action dated Oct. 15, 2020 in U.S. Appl. No. 15/925,532. cited by applicant

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USPTO; Non-Final Office Action dated Jul. 22, 2019 in U.S. Appl. No. 15/940,759. cited by applicant

USPTO; Final Office Action dated Feb. 19, 2020 in U.S. Appl. No. 15/940,759. cited by applicant

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USPTO; Non-Final Office Action dated Aug. 3, 2020 in U.S. Appl. No. 15/940,759. cited by applicant

USPTO; Non-Final Office Action dated Mar. 29, 2019 in U.S. Appl. No. 15/940,801. cited by applicant

USPTO; Notice of Allowance dated Aug. 26, 2019 in U.S. Appl. No. 15/940,801. cited by applicant

USPTO; Non-Final Rejection dated Nov. 29, 2019 in U.S. Appl. No. 15/949,990. cited by applicant

USPTO; Notice of Allowance dated Mar. 12, 2020 in U.S. Appl. No. 15/949,990. cited by applicant

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USPTO; Non-Final Office Action dated Jan. 31, 2020 in U.S. Appl. No. 15/962,980. cited by applicant

USPTO; Final Office Action dated Apr. 30, 2020 in U.S. Appl. No. 15/962,980. cited by applicant

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USPTO; Non-Final Office Action dated Sep. 14, 2020 in U.S. Appl. No. 15/962,980. cited by applicant

USPTO; Non-Final Office Action dated Oct. 6, 2020 in U.S. Appl. No. 15/967,146. cited by applicant

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USPTO; Final Office Action dated Sep. 3, 2020 in U.S. Appl. No. 15/974,948. cited by applicant

USPTO; Non-Final Office Action dated May 15, 2020 in U.S. Appl. No. 15/974,988. cited by applicant

USPTO; Notice of Allowance dated Oct. 14, 2020 in U.S. Appl. No. 15/974,988. cited by applicant

USPTO; Non-Final Office Action dated Jan. 16, 2020 in U.S. Appl. No. 15/985,261. cited by applicant

USPTO; Notice of Allowance dated Apr. 3, 2020 in U.S. Appl. No. 15/985,261. cited by applicant

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USPTO; Final Office Action dated Nov. 25, 2020 in U.S. Appl. No. 15/985,539. cited by applicant

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USPTO; Non-Final Office Action dated Apr. 22, 2020 in U.S. Appl. No. 16/000,156. cited by applicant
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USPTO; Non-Final Office Action dated Jul. 16, 2019 in U.S. Appl. No. 16/014,981. cited by applicant
USPTO; Non-Final Office Action dated Jan. 16, 2020 in U.S. Appl. No. 16/014,981. cited by applicant
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USPTO; Notice of Allowance dated Apr. 9, 2019 in U.S. Appl. No. 16/026,711. cited by applicant
USPTO; Non-Final Office Action dated Mar. 4, 2020 in U.S. Appl. No. 16/036,692. cited by applicant
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USPTO; Advisory Action dated Nov. 17, 2020 in U.S. Appl. No. 16/116,708. cited by applicant

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USPTO; Non-Final Office Action dated Jan. 6, 2020 in U.S. Appl. No. 16/132,142. cited by applicant

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applicant

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USPTO; Non-Final Office Action dated Nov. 9, 2020 in U.S. Appl. No. 16/152,260. cited by

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USPTO; Final Office Action dated Mar. 2, 2020 in U.S. Appl. No. 16/158,077. cited by applicant

USPTO; Advisory Action dated Jun. 9, 2020 in U.S. Appl. No. 16/158,077. cited by applicant

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USPTO; Notice of Allowance dated Jun. 19, 2020 in U.S. Appl. No. 16/161,744. cited by applicant

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applicant

USPTO; Non-Final Office Action dated Aug. 16, 2019 in U.S. Appl. No. 16/167,225. cited by

applicant

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USPTO; Notice of Allowance dated Mar. 5, 2020 in U.S. Appl. No. 16/167,225. cited by applicant

USPTO; Notice of Allowance dated Apr. 17, 2019 in U.S. Appl. No. 16/171,098. cited by applicant

USPTO; Notice of Allowance dated May 1, 2019 in U.S. Appl. No. 16/171,098. cited by applicant

USPTO; Non-Final Office Action dated Aug. 7, 2020 in U.S. Appl. No. 16/172,535. cited by

applicant

USPTO; Final Office Action dated Oct. 16, 2020 in U.S. Appl. No. 16/172,535. cited by applicant

USPTO; Non-Final Office Action dated Nov. 4, 2020 in U.S. Appl. No. 16/176,517. cited by

applicant

USPTO; Non-Final Office Action dated Oct. 29, 2020 in U.S. Appl. No. 16/183,258. cited by

applicant

USPTO; Non-Final Office Action dated Apr. 2, 2019 in U.S. Appl. No. 16/188,690. cited by

applicant

USPTO; Final Office Action dated Sep. 26, 2019 in U.S. Appl. No. 16/188,690. cited by applicant

USPTO; Notice of Allowance dated Dec. 16, 2019 in U.S. Appl. No. 16/188,690. cited by applicant

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USPTO; Notice of Allowance dated Oct. 3, 2019 in U.S. Appl. No. 16/200,100. cited by applicant

USPTO; Non-Final Office Action dated May 29, 2020 in U.S. Appl. No. 16/205,899. cited by

applicant

USPTO; Final Office Action dated Nov. 27, 2020 in U.S. Appl. No. 16/205,899. cited by applicant

USPTO; Non-Final Office Action dated Aug. 19, 2019 in U.S. Appl. No. 16/208,062. cited by

applicant

USPTO; Notice of Allowance dated Jan. 28, 2020 in U.S. Appl. No. 16/208,062. cited by applicant

USPTO; Non-Final Office Action dated Jul. 15, 2020 in U.S. Appl. No. 16/210,922. cited by

applicant

USPTO; Non-Final Office Action dated Sep. 16, 2019 in U.S. Appl. No. 16/213,702. cited by

applicant

USPTO; Notice of Allowance dated Dec. 9, 2019 in U.S. Appl. No. 16/213,702. cited by applicant

USPTO; Non-Final Office Action dated Jan. 8, 2020 in U.S. Appl. No. 16/219,555. cited by

applicant

USPTO; Final Office Action dated Jun. 19, 2020 in U.S. Appl. No. 16/219,555. cited by applicant

USPTO; Advisory Action dated Sep. 11, 2020 in U.S. Appl. No. 16/219,555. cited by applicant
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USPTO; Non-Final Office Action dated Feb. 19, 2020 in U.S. Appl. No. 16/242,829. cited by applicant
USPTO; Final Office Action dated Jun. 15, 2020 in U.S. Appl. No. 16/242,829. cited by applicant
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USPTO; Non-Final Office Action dated Jan. 30, 2020 in U.S. Appl. No. 16/245,006. cited by applicant
USPTO; Notice of Allowance dated May 29, 2020 in U.S. Appl. No. 16/245,006. cited by applicant
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USPTO; Final Office Action dated Jan. 10, 2020 in U.S. Appl. No. 16/251,534. cited by applicant
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USPTO; Non-Final Office Action dated May 27, 2020 in U.S. Appl. No. 16/252,567. cited by applicant
USPTO; Final Office Action dated Nov. 18, 2020 in U.S. Appl. No. 16/252,567. cited by applicant
USPTO; Non-Final Office Action dated Apr. 16, 2020 in U.S. Appl. No. 16/255,639. cited by applicant
USPTO; Notice of Allowance dated Aug. 6, 2020 in U.S. Appl. No. 16/255,639. cited by applicant
USPTO; Non-Final Office Action dated Jun. 30, 2020 in U.S. Appl. No. 16/258,187. cited by applicant
USPTO; Notice of Allowance dated Aug. 5, 2020 in U.S. Appl. No. 16/258,187. cited by applicant
USPTO; Non-Final Office Action dated Dec. 18, 2019 in U.S. Appl. No. 16/280,964. cited by applicant
USPTO; Notice of Allowance dated May 21, 2020 in U.S. Appl. No. 16/280,964. cited by applicant
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USPTO; Notice of Allowance dated Aug. 31, 2020 in U.S. Appl. No. 16/356,394. cited by applicant
USPTO; Notice of Allowance dated Jun. 13, 2019 in U.S. Appl. No. 16/396,475. cited by applicant
USPTO; Non-Final Office Action dated Aug. 5, 2020 in U.S. Appl. No. 16/397,045. cited by applicant
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USPTO; Notice of Allowance dated Mar. 30, 2020 in U.S. Appl. No. 16/417,938. cited by applicant
USPTO; Non-Final Office Action dated Dec. 31, 2019 in U.S. Appl. No. 16/427,288. cited by applicant
USPTO; Notice of Allowance dated Apr. 16, 2020 in U.S. Appl. No. 16/427,288. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a non-provisional of, and claims priority to and the benefit of, U.S. Provisional Patent Application No. 62/967,320, filed Jan. 29, 2020 and entitled "CONTAMINANT TRAP SYSTEM FOR A REACTOR SYSTEM," which is hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to a semiconductor processing or reactor system and components comprised therein, and particularly to reactor system components that prevent contamination of other components.

BACKGROUND OF THE DISCLOSURE

(2) Reaction chambers may be used for depositing various material layers onto semiconductor substrates. A semiconductor may be placed on a susceptor inside a reaction chamber. Both the substrate and the susceptor may be heated to a desired substrate temperature set point. In an example substrate treatment process, one or more reactant gases may be passed over a heated substrate, causing the deposition of a thin film of material on the substrate surface. Throughout subsequent deposition, doping, lithography, etch, and other processes, these layers are made into integrated circuits.

(3) For any given process, reactant gases and/or any byproduct gases may then be evacuated via a vacuum and/or purged from the reaction chamber. Reactant gases, and other gases or materials from the reaction chamber may be passed through a filter or a contaminant trap system, wherein the reactant gases or other materials (e.g., reaction products and/or byproducts) are trapped to prevent contamination of reactor system components downstream of the contaminant trap system. However, materials from the contaminant trap system may outgas under certain conditions, which may cause contamination of the reaction chamber or a substrate disposed therein.

SUMMARY OF THE DISCLOSURE

(4) This summary is provided to introduce a selection of concepts in a simplified form. These concepts are described in further detail in the detailed description of example embodiments of the disclosure below. This summary is not intended to necessarily identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(5) In some embodiments, a contaminant trap system for a reactor system is provided. The contaminant trap system disclosed herein may allow collection of materials from a reaction chamber of the reactor system to reduce or prevent contamination of reactor system components downstream of the contaminant trap system. The contaminant trap system disclosed herein may also reduce or prevent possible contaminants from traveling to and contaminating the reaction chamber or a substrate disposed therein.

(6) In various embodiments, a baffle plate stack for a contaminant trap system may comprise a plurality of baffle plates, each comprising an aperture spanning through a baffle plate body of each baffle plate of the plurality of baffle plates, and a solid body portion; and a plurality of complementary baffle plates, each comprising a complementary aperture spanning through a complementary baffle plate body of each complementary baffle plate of the plurality of complementary baffle plates, and a complementary solid body portion. The plurality of baffle plates and the plurality of complementary baffle plates may be disposed in a baffle plate order between a first end and a second end of the baffle plate stack in which the plurality of baffle plates alternates with the plurality of complementary baffle plates, such that no two of the plurality of baffle plates and no two of the plurality of complementary baffle plates are adjacent in the baffle plate order. The plurality of baffle plates and the plurality of complementary baffle plates may be disposed in a baffle plate orientation wherein at least a portion of the apertures of the plurality of baffle plates and at least a portion of the complementary solid body portions of the plurality of complementary baffle plates may be aligned along a first axis spanning between the first end and the second end of the baffle plate stack, and such that at least a portion of the solid body portions of the plurality of baffle plates and at least a portion of the complementary apertures of the plurality of complementary baffle plates may be aligned along a second axis spanning between the first end and the second end of the baffle plate stack.

(7) In various embodiments, a baffle plate stack may further comprise a coupling rod coupled to each of the plurality of baffle plates and/or each of the plurality of complementary baffle plates, wherein the coupling rod may span between the first end and the second end of the baffle plate stack, wherein the coupling rod comprises a cross-section. Each of the plurality of baffle plates may comprise a coupling hole and each of the plurality of complementary baffle plates may comprise a complementary coupling hole, wherein the coupling holes and the complementary coupling holes each may comprise a shape complementary to the cross-section of the coupling rod. In various embodiments, the cross-section of the coupling rod may be noncircular, wherein the coupling hole of each of the plurality of baffle plates may be disposed in a first orientation, and the complementary coupling hole of each of the plurality of complementary baffle plates may be disposed in a second orientation. The first orientation and the second orientation may dispose the plurality of baffle plates and the plurality of complementary baffle plates about the coupling rod to achieve the baffle plate orientation.

(8) In various embodiments, a baffle plate stack may further comprise a plurality of spacers coupled to the coupling rod, wherein at least one of the plurality of spacers may be disposed between each baffle plate and complementary baffle plate of the plurality of baffle plates and the plurality of complementary baffle plates in the baffle plate order. In various embodiments, a baffle plate stack may further comprise an end plate disposed at least one of at the first end or the second end of the baffle plate stack, wherein the end plate may comprise an end plate aperture and an end plate solid body portion.

(9) In various embodiments, there may be one more of the plurality of baffle plates than the plurality of complementary baffle plates, such that baffle plate stack may comprise the same order of the plurality of baffle plates and the plurality of complementary baffle plates from the first end and the second end of the baffle plate stack. In various embodiments, at least one of the plurality of baffle plates and the plurality of complementary baffle plates may comprise a textured surface.

(10) In various embodiments, a contaminant trap system of a reactor system may comprise a trap housing comprising a housing outer wall; a first baffle plate disposed in the trap housing, wherein the first baffle plate may comprise a first aperture spanning through a first baffle plate body between a first top baffle plate surface and a first bottom baffle plate surface of the first baffle plate, and a first solid body portion; a first complementary baffle plate disposed in the trap housing in series with the first baffle plate between a first end and a second end of the trap housing, wherein the first complementary baffle plate may comprise a first complementary aperture spanning through a first complementary baffle plate body between a first top complementary baffle plate surface and a first bottom complementary baffle plate surface of the first complementary baffle plate, and a first complementary solid body portion. The first baffle plate and the first complementary baffle plate may be comprised in a baffle plate stack. The first baffle plate and the first complementary baffle plate may be disposed in a baffle plate orientation in the trap housing wherein at least a portion of the first aperture of the first baffle plate and at least a portion of the first complementary solid body portion of the first complementary baffle plate may be aligned along a first axis spanning between the first end and the second end of the trap housing, and such that at least a portion of the first solid body portion of the first baffle plate and at least a portion of the first complementary aperture of the first complementary baffle plate may be aligned along a second axis spanning between the first end and the second end of the trap housing. In various embodiments, the first aperture of the first baffle plate may be comprised in a radially inward portion of the first baffle plate, and/or the first complementary aperture of the first complementary baffle plate may be comprised in a radially outward portion of the first complementary baffle plate. In various embodiments, the contaminant trap system may further comprise a heater jacket coupled to the trap housing.

(11) In various embodiments, the contaminant trap system may further comprise a coupling rod disposed in the trap housing and spanning between the first end and the second end of the trap

housing. The first baffle plate may comprise a first coupling hole disposed through the first baffle plate body, wherein the coupling rod may be disposed through the first coupling hole. The first complementary baffle plate may comprise a first complementary coupling hole disposed through the first complementary baffle plate body, wherein the coupling rod may be disposed through the first complementary coupling hole. In various embodiments, the coupling rod may comprise a noncircular cross-section, wherein the first coupling hole of the first baffle plate and the first complementary coupling hole of the first complementary baffle plate each may comprise a shape complementary to the noncircular cross-section of the coupling rod. In various embodiments, a reference point of the first coupling hole may be disposed in a first orientation, and a complementary reference point of the first complementary coupling hole may be disposed in a first complementary orientation, wherein the first orientation and the first complementary orientation may dispose the first baffle plate and the first complementary baffle plate about the coupling rod to achieve the baffle plate orientation.

(12) In various embodiments, the contaminant trap system may further comprise a spacer between the first baffle plate and the first complementary baffle plate to provide a space therebetween.

(13) In various embodiments, the contaminant trap system may further comprise a second baffle plate disposed in the trap housing, wherein the second baffle plate may comprise a second aperture spanning through a second baffle plate body between a second top baffle plate surface and a second bottom baffle plate surface of the second baffle plate, and a second solid body portion. The second baffle plate may be disposed in the trap housing such that the first complementary baffle plate may be between the first baffle plate and the second baffle plate, and wherein the baffle plate orientation may further comprise at least a portion of the second aperture of the second baffle plate and at least a portion of the first complementary solid body portion of the first complementary baffle plate being aligned along the first axis, and such that at least a portion of the second solid body portion of the second baffle plate and at least a portion of the first complementary aperture of the first complementary baffle plate may be aligned along the second axis. In various embodiments, the first baffle plate and the second baffle plate may comprise an identical design.

(14) In various embodiments, the baffle plate stack may further comprise an end plate disposed such that the first baffle plate is between the end plate and the first complementary baffle plate, or the first complementary baffle plate is between the end plate and the first baffle plate. The end plate may comprise an end plate aperture and an end plate solid body portion.

(15) In various embodiments, the housing outer wall of the trap housing may comprise an interior wall surface. An outer edge of at least one of the first baffle plate and the first complementary baffle plate may be disposed adjacent to the interior wall surface such that at least a partial seal may be formed between the outer edge of the first baffle plate and/or the first complementary baffle plate, and the interior wall surface.

(16) In various embodiments, the first top baffle plate surface, the first bottom baffle plate surface, the first top complementary baffle plate surface, the first bottom complementary baffle plate surface, the outer edge of at least one of the first baffle plate and the first complementary baffle plate, and/or the interior wall surface is textured.

(17) In various embodiments, a method may comprise flowing a fluid from a reaction chamber into a trap housing of a contaminant trap system; flowing the fluid through a baffle plate stack disposed in the trap housing and comprising a plurality of baffle plates and a plurality of complementary baffle plates; flowing the fluid through an aperture of a first baffle plate of the plurality of baffle plates; flowing the fluid into the complementary solid body portion of a first complementary baffle plate of the plurality of complementary baffle plates in response to the flowing the fluid through the aperture of the first baffle plate; depositing contaminants onto the complementary solid body portion of the first complementary baffle plate in response to the flowing the fluid into the complementary solid body portion of the first complementary baffle plate; flowing the fluid through a complementary aperture of the first complementary baffle plate in response to the

flowing the fluid into the complementary solid body portion of the first complementary baffle plate; flowing the fluid into the solid body portion of a second baffle plate of the plurality of baffle plates in response to the flowing the fluid through the complementary aperture of the first complementary baffle plate; and/or depositing contaminants onto the solid body portion of the second baffle plate in response to the flowing the fluid into the solid body portion of the second baffle plate. Each of the plurality of baffle plates may comprise a solid body portion and an aperture spanning through a baffle plate body of each baffle plate of the plurality of baffle plates. Each of the plurality of complementary baffle plates may comprise a complementary solid body portion and a complementary aperture spanning through a complementary baffle plate body of each complementary baffle plate of the plurality of complementary baffle plates. The plurality of baffle plates and the plurality of complementary baffle plates may be disposed in a baffle plate order between a first end and a second end of the baffle plate stack in which the plurality of baffle plates may alternate with the plurality of complementary baffle plates, such that no two of the plurality of baffle plates and no two of the plurality of complementary baffle plates are adjacent in the baffle plate order. The plurality of baffle plates and the plurality of complementary baffle plates may be disposed in a baffle plate orientation wherein at least a portion of the apertures of the plurality of baffle plates and at least a portion of the complementary solid body portions of the plurality of complementary baffle plates may be aligned along a first axis spanning between the first end and the second end of the baffle plate stack, and such that at least a portion of the solid body portions of the plurality of baffle plates and at least a portion of the complementary apertures of the plurality of complementary baffle plates may be aligned along a second axis spanning between the first end and the second end of the baffle plate stack.

(18) In various embodiments, a contaminant trap system of a reactor system may comprise a trap housing; and a trap structure disposed in the trap housing. The trap structure may comprise a baffle plate; a base plate; and a plurality of rods spanning between and coupled to the baffle plate and the base plate. The rods may be disposed about a flow hole disposed through the base plate.

(19) In various embodiments, a contaminant trap system of a reactor system may comprise a trap housing comprising a housing bottom surface and a housing top surface; and a trap structure disposed in the trap housing. The trap structure may comprise a plurality of tubes disposed in an arrangement having an outer shape complementary to a shape of the trap housing; a support disposed within the arrangement of the plurality of tubes and protruding outwardly from an end of the plurality of tubes, wherein the support contacts the housing bottom surface, creating a space between the end of the plurality of tubes and the housing bottom surface; and a tensioning device coupled around the plurality of tubes configured to hold the plurality of tubes together. The plurality of tubes may be packed hexagonally, wherein each tube of the plurality of tubes comprises a bore and may span at least partially between the housing bottom surface and the housing top surface.

(20) In various embodiments, a contaminant trap of a reactor system may comprise a trap housing; and a trap structure disposed in the trap housing. The trap structure may comprise a corrugated sheet coupled to a noncorrugated sheet. The corrugated and noncorrugated sheets may be spiraled such that portions of the corrugated sheet are disposed between portions of the noncorrugated sheet, and such that portions of the noncorrugated sheet are disposed between portions of the corrugated sheet.

(21) For the purpose of summarizing the disclosure and the advantages achieved over the prior art, certain objects and advantages of the disclosure have been described herein above. Of course, it is to be understood that not necessarily all such objects or advantages may be achieved in accordance with any particular embodiment of the disclosure. Thus, for example, those skilled in the art will recognize that the embodiments disclosed herein may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught or suggested herein without necessarily achieving other objects or advantages as may be taught or suggested herein.

(22) All of these embodiments are intended to be within the scope of the disclosure. These and other embodiments will become readily apparent to those skilled in the art from the following detailed description of certain embodiments having reference to the attached figures, the disclosure not being limited to any particular embodiment(s) discussed.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) While the specification concludes with claims particularly pointing out and distinctly claiming what are regarded as embodiments of the disclosure, the advantages of embodiments of the disclosure may be more readily ascertained from the description of certain examples of the embodiments of the disclosure when read in conjunction with the accompanying drawings. Elements with the like element numbering throughout the figures are intended to be the same.
- (2) FIG. 1 illustrates a schematic diagram of an exemplary reactor system, in accordance with various embodiments;
- (3) FIG. 2 illustrates an exploded view of an exemplary contaminant trap system, in accordance with various embodiments;
- (4) FIG. 3A illustrates an exemplary baffle plate, in accordance with various embodiments;
- (5) FIG. 3B illustrates an exemplary complementary baffle plate, in accordance with various embodiments;
- (6) FIG. 4A illustrates a perspective view of an exemplary filter baffle plate stack of a contaminant trap system, in accordance with various embodiments;
- (7) FIG. 4B illustrates a cross-sectional perspective view of an exemplary filter baffle plate stack of a contaminant trap system, in accordance with various embodiments;
- (8) FIG. 5A illustrates another exemplary baffle plate, in accordance with various embodiments;
- (9) FIG. 5B illustrates another exemplary complementary baffle plate, in accordance with various embodiments;
- (10) FIG. 6A illustrates yet another exemplary baffle plate, in accordance with various embodiments;
- (11) FIG. 6B illustrates yet another exemplary complementary baffle plate, in accordance with various embodiments;
- (12) FIG. 7A illustrates yet another exemplary baffle plate, in accordance with various embodiments;
- (13) FIG. 7B illustrates yet another exemplary complementary baffle plate, in accordance with various embodiments;
- (14) FIG. 8 illustrates a perspective view of a heater jacket for a contaminant trap system, in accordance with various embodiments;
- (15) FIG. 9 illustrates a method of flowing fluid through a contaminant trap system of a reactor system, in accordance with various embodiments;
- (16) FIG. 10A illustrates a cross-sectional view of an exemplary trap structure, in accordance with various embodiments;
- (17) FIG. 10B illustrates an exploded view of the cross section of the trap structure of FIG. 10A, in accordance with various embodiments;
- (18) FIG. 11 illustrates an exemplary trap structure, in accordance with various embodiments; and
- (19) FIG. 12 illustrates an exemplary trap structure, in accordance with various embodiments.

DETAILED DESCRIPTION

(20) Although certain embodiments and examples are disclosed below, it will be understood by those in the art that the disclosure extends beyond the specifically disclosed embodiments and/or uses of the disclosure and obvious modifications and equivalents thereof. Thus, it is intended that

the scope of the disclosure should not be limited by the particular embodiments described herein. (21) The illustrations presented herein are not meant to be actual views of any particular material, apparatus, structure, or device, but are merely representations that are used to describe embodiments of the disclosure.

(22) As used herein, the term “substrate” may refer to any underlying material or materials that may be used, or upon which, a device, a circuit, or a film may be formed.

(23) As used herein, the term “atomic layer deposition” (ALD) may refer to a vapor deposition process in which deposition cycles, preferably a plurality of consecutive deposition cycles, are conducted in a process chamber. Typically, during each cycle the precursor is chemisorbed to a deposition surface (e.g., a substrate surface or a previously deposited underlying surface such as material from a previous ALD cycle), forming a monolayer or sub-monolayer that does not readily react with additional precursor (i.e., a self-limiting reaction). Thereafter, if necessary, a reactant (e.g., another precursor or reaction gas) may subsequently be introduced into the process chamber for use in converting the chemisorbed precursor to the desired material on the deposition surface. Typically, this reactant is capable of further reaction with the precursor. Further, purging steps may also be utilized during each cycle to remove excess precursor from the process chamber and/or remove excess reactant and/or reaction byproducts from the process chamber after conversion of the chemisorbed precursor. Further, the term “atomic layer deposition,” as used herein, is also meant to include processes designated by related terms such as, “chemical vapor atomic layer deposition”, “atomic layer epitaxy” (ALE), molecular beam epitaxy (MBE), gas source MBE, or organometallic MBE, and chemical beam epitaxy when performed with alternating pulses of precursor composition(s), reactive gas, and purge (e.g., inert carrier) gas.

(24) As used herein, the term “chemical vapor deposition” (CVD) may refer to any process wherein a substrate is exposed to one or more volatile precursors, which react and/or decompose on a substrate surface to produce a desired deposition.

(25) As used herein, the term “film” and “thin film” may refer to any continuous or non-continuous structures and material deposited by the methods disclosed herein. For example, “film” and “thin film” could include 2D materials, nanorods, nanotubes, or nanoparticles or even partial or full molecular layers or partial or full atomic layers or clusters of atoms and/or molecules. “Film” and “thin film” may comprise material or a layer with pinholes, but still be at least partially continuous.

(26) As used herein, the term “contaminant” may refer to any unwanted material disposed within the reaction chamber that may affect the purity of a substrate disposed in the reaction chamber, or any unwanted material in any component of a reaction system. The term “contaminant” may refer to, but is not limited to, unwanted deposits, metal and non-metal particles, impurities, and waste products, disposed within the reaction chamber or other components of the reactor system.

(27) Reactor systems used for ALD, CVD, and/or the like, may be used for a variety of applications, including depositing and etching materials on a substrate surface. In various embodiments, a reactor system **50** may comprise a reaction chamber **4**, a susceptor **6** to hold a substrate **30** during processing, a fluid distribution system **8** (e.g., a showerhead) to distribute one or more reactants to a surface of substrate **30**, one or more reactant sources **10**, **12**, and/or a carrier and/or purge gas source **14**, fluidly coupled to reaction chamber **4** via lines **16-20**, and valves or controllers **22-26**. Reactant gases or other materials from reactant sources **10**, **12** may be applied to substrate **30** in reaction chamber **4**. A purge gas from purge gas source **14** may be flowed to and through reaction chamber **4** to remove any excess reactant or other undesired materials from reaction chamber **4**. System **50** may also comprise a vacuum source **28** fluidly coupled to the reaction chamber **4**, which may be configured to suck reactants, a purge gas, or other materials out of reaction chamber **4**. System **50** may comprise a contaminant trap system **40** disposed between reaction chamber **4** and vacuum source **28** to trap (i.e., accumulate) materials (e.g., contaminants) coming from reaction chamber **4**, reducing or preventing contamination of reactor system **50** components downstream of contaminant trap system **40**.

(28) With reference to FIG. 2, a contaminant trap system **100** (an exploded view thereof) is illustrated (an example of contaminant trap system **40** in FIG. 1), in accordance with various embodiments. In various embodiments, contaminant trap system **100** may comprise a trap housing **103**, which may comprise multiple components (e.g., upper housing **103A** and lower housing **103B**). In various embodiments, upper housing **103A** and lower housing **103B** may couple to enclose other components of contaminant trap system **100**. Upper housing **103A** may comprise a fluid inlet **101A**, through which the interior of trap housing **103** may be fluidly coupled to a reaction chamber (e.g., reaction chamber **4**). Gases and other material may flow from the reaction chamber into contaminant trap system **100** through fluid inlet **101A**, and may exit contaminant trap system **100** through fluid outlet **101B**.

(29) In various embodiments, trap housing **103** may comprise an outer wall **105** comprising an interior wall surface. The interior wall surface may define the internal space enclosed within trap housing **103** (e.g., when upper housing **103A** and lower housing **103B** are coupled). The internal space of trap housing **103** may be in fluid communication with fluid inlet **101A** and fluid outlet **101B**.

(30) In various embodiments, contaminant trap system **100** may comprise a trap structure (e.g., housed within the trap housing) configured to trap contaminants traveling therethrough. The contaminants may deposit on the surfaces of the trap structure as fluid travels through the trap system. In various embodiments, the trap structure may comprise a baffle plate stack (e.g., baffle plate stack **130**). Baffle plate stack **130** may comprise at least two plates, which may cause fluid flow entering the internal space of trap housing **103** to take a certain path (e.g., a path that will increase or maximize fluid flow through the internal space of trap housing **103**, and/or allow increased or maximum removal of contaminants from the fluid flow by the contaminant trap system and its components). The fluid flow path through the internal space of trap housing **103** may increase the fluid path to increase contact with the components of contaminant trap system **100** (e.g., the surfaces of the baffle plates in trap housing **103**), therefore, allowing more opportunities for contaminant deposition on such surfaces as the fluid flows through contaminant trap system **100**.

(31) In various embodiments, baffle plate stack **130** may comprise at least one baffle plate **132** and at least one complementary baffle plate **134**. Each baffle plate **132** may have substantially the same design (e.g., comprising apertures therethrough), and each complementary baffle plate **134** may have substantially the same design (e.g., comprising complementary apertures therethrough). Baffle plates **132** and complementary baffle plates **134** may be disposed in a baffle plate order in baffle plate stack **130** between a first end of **103** (proximate **101A**) and a second end of trap housing **103** (proximate fluid outlet **101B**). A first end of baffle plate stack **130** may be proximate the first end of trap housing **103**, and a second end of baffle plate stack **130** may be proximate the second end of trap housing **103**. The baffle plate order may comprise baffle plates **132** alternating positions with complementary baffle plates **134**, such that no two baffle plates **132**, and no two complementary baffle plates **134**, are adjacent in the baffle plate order.

(32) Baffle plate stack **130** may comprise any suitable number of baffle plates of any design, order, and/or composition. For example, in various embodiments, baffle plate stack **130** may comprise all of one type of baffle plate (e.g., all of baffle plates **132** or all of complementary baffle plates **134**). In various embodiments, baffle plate stack **130** may comprise any suitable mixture of baffle plate designs. For example, baffle plate stack **130** may comprise baffle plates comprising two or more designs. As a further example, baffle plate stack **130** may comprise a first number of baffle plates **132** and a second number of complementary baffle plates **134**. In various embodiments, baffle plate stack **130** may comprise an equal number of baffle plates **132** and complementary baffle plates **134** (e.g., alternating in the baffle plate order between the first and second ends of baffle plate stack **130**). In various embodiments, baffle plate stack **130** may comprise one more baffle plate **132** than complementary baffle plates **134**, such that the baffle plate order starts and ends with a baffle plate

132 (i.e., a baffle plate **132** is the baffle plate most proximate the first and second ends of trap housing **103**).

(33) In various embodiments, the baffle plate stack may comprise at least one end plate coupled to each end of the baffle plate stack. For example, a first end plate **136A** may be comprised in baffle plate stack **130** as the end plate on a first end of baffle plate stack **130**, and a second end plate **136B** may be comprised in baffle plate stack **130** as the end plate on a second end of baffle plate stack **130**. The first end of baffle plate stack **130** may be disposed in the internal space of trap housing **103** proximate the first end of trap housing **103**, and the second end of baffle plate stack **130** may be disposed in the internal space of trap housing **103** proximate the second end of trap housing **103**. The end plate(s) comprised in a baffle plate stack may comprise any suitable design, including a design different than the baffle plates and/or complementary baffle plates comprised in the baffle plate stack.

(34) The arrangement of the baffle plates in baffle plate stack **130** may comprise any suitable arrangement, including any suitable spacing arrangement. The baffle plates may each be separated by a spacer **133**. That is, a spacer **133** may be disposed between every two plates in the baffle plate stack. The plates in the baffle plate stack may be spaced apart by any suitable distance, for example, to accomplish a desired pressure drop of the fluid flow through trap housing **103**. To decrease the amount of pressure drop through trap housing **103**, there may be fewer baffle plates in the baffle plate stack, and/or more space between the baffle plates. Conversely, to increase the amount of pressure drop through trap housing **103**, there may be more baffle plates in the baffle plate stack, and/or less space between the baffle plates.

(35) Each baffle plate (e.g., baffle plates **132** and complementary baffle plates **134** in baffle plate stack **130**) may comprise a shape that is complementary to the internal space of trap housing **103** such that baffle plate stack **130** and the baffle plates comprised therein may be disposed in the internal space of trap housing **103**. In various embodiments, an outer edge of one or more plates comprised in the baffle plate stack disposed in the internal space of trap housing **103** may be disposed adjacent to and/or in contact with the interior wall of trap housing **103**. The outer edge of one or more of the plates may form at least a partial seal between the respective baffle plate and the interior wall of trap housing **103**. Therefore, a limited amount of fluid flow (or no fluid flow) may pass between the outer edges of the plates in a baffle plate stack and the interior wall of trap housing **103**.

(36) With reference to FIGS. 3A, 3B and 4A, in various embodiments, a baffle plate (e.g., baffle plate **300A**, which is an example of baffle plate **132** in FIG. 2) may comprise a top surface **322**, a bottom surface **324**, a baffle plate body therebetween, and a baffle plate outer edge **326**. A baffle plate may comprise at least one aperture disposed through the baffle plate body between top surface **322** and bottom surface **324** and defined by an aperture edge. For example, baffle plate **300A** may comprise first apertures **331** and second apertures **333**. The apertures comprised in a baffle plate may be in any suitable aperture arrangement, such as the aperture arrangement of baffle plate **300A** shown in FIG. 3A. As an example of an aperture arrangement of a baffle plate, the apertures may be spaced equidistant from other like apertures about a center of the baffle plate shape (e.g., the center of a circle). In various embodiments, the apertures of a baffle plate may be comprised in an aperture portion of the baffle plate. For example, aperture portion **325** of baffle plate **300A** may be disposed on a radially-inward portion of the baffle plate, wherein the radially-outward portion of baffle plate **300A** may not comprise an aperture. The portion of a baffle plate without an aperture may be a solid body portion (e.g., solid body portion **335** of baffle plate **300A**).

(37) With continued reference to FIGS. 3A, 3B and 4A, in various embodiments, a complementary baffle plate (e.g., complementary baffle plate **300B**, which is an example of complementary baffle plate **134** in FIG. 2) may comprise a complementary top surface **352**, a complementary bottom surface **354**, a complementary baffle plate body therebetween, and a complementary baffle plate outer edge **356**. A complementary baffle plate may comprise at least one complementary aperture

disposed through the complementary baffle plate body between complementary baffle plate top surface **352** and bottom surface **354** and defined by a complementary aperture edge. For example, complementary baffle plate **300B** may comprise first complementary apertures **361** and second complementary apertures **363**. The complementary apertures comprised in a complementary baffle plate may be in any suitable complementary aperture arrangement, such as the complementary aperture arrangement of complementary baffle plate **300B** shown in FIG. **3B**. As an example of a complementary aperture arrangement of a complementary baffle plate, the complementary apertures may be spaced equidistant from other like complementary apertures about a center of the complementary baffle plate shape (e.g., the center of a circle). In various embodiments, the complementary apertures of a complementary baffle plate may be comprised in a complementary aperture portion of the complementary baffle plate. For example, aperture portion **355** of complementary baffle plate **300B** may be disposed on a radially-outward portion of the complementary baffle plate, wherein the radially-inward portion of complementary baffle plate **300B** may not comprise a complementary aperture. The portion of a complementary baffle plate without a complementary aperture may be a complementary solid body portion (e.g., complementary solid body portion **365** of complementary baffle plate **300B**).

(38) Complementary baffle plates (e.g., complementary baffle plates **300B**) may be complementary to baffle plates (e.g., baffle plates **30A**) because complementary baffle plates may comprise complementary apertures in portions of the plate which the baffle plates do not comprise apertures. As an example, discussed above, complementary baffle plates **300B** comprise complementary apertures **361** and **363** in a radially outward portion thereof, while baffle plates **300A** do not comprise apertures in a radially outward portion thereof.

(39) In various embodiments, a baffle plate stack may comprise a coupling rod to which baffle plates and/or complementary baffle plates may couple. For example, baffle plate stack **400B** in FIG. **4B** may comprise coupling rod **450**. The coupling rod may comprise any suitable shape, length, and/or cross-sectional shape. In various embodiments, the coupling rod may be configured to span between the first and second ends of trap housing **103**. The coupling rod may be configured to engage and/or couple with other components of a baffle plate stack, such as baffle plates, complementary baffle plates, end plates, spacers, and/or the like. In various embodiments, at least a portion of a coupling rod may comprise threading, such as one or more of the ends of coupling rod **450** to engage with a fastener to secure the baffle plates, complementary baffle plates, end plates, and/or spacers together.

(40) For space and clarity purposes, the reference numbers and lead lines for specific baffle plate components and complementary baffle plate components in FIGS. **4A** and **4B** are included in one or more exemplary baffle plates or complementary baffle plates illustrated therein. However, such labeled components may apply to each similarly labeled baffle plate or complementary baffle plate, as appropriate.

(41) In various embodiments, each baffle plate may comprise a coupling hole configured to receive and/or engage with the coupling rod. For example, baffle plate **300A** may comprise a coupling hole **347** having a shape that is complementary to a cross-sectional shape of coupling rod **450**. Therefore, coupling rod **450** may be inserted through coupling hole **347**, and coupling hole **347** may engage with coupling rod **450**.

(42) In various embodiments, the coupling hole of a baffle plate may comprise a non-circular shape, such that the coupling rod may engage with the coupling hole and maintain the baffle plate a desired position (e.g., so baffle plate **300A** does not rotate about coupling rod **450** within trap housing **103**). In various embodiments, the coupling hole of a baffle plate may comprise a shape that is symmetrical about only one line passing through the coupling hole (e.g., through a center of the coupling hole). That way, the coupling hole may only engage with the coupling rod in a way that disposes the baffle plate in a desired orientation (a self-aligning feature). In various embodiments, to aid in disposing a baffle plate in a desired orientation about the coupling rod, the

coupling hole may comprise a reference point which is disposed in a specific orientation or at a specific angle, and/or a specific position relative to the aperture(s) of the baffle plate. For example, coupling hole **347** may comprise reference point **348** which may be oriented at a specific angle (e.g., such that reference point **348** is aligned with a first aperture **331** and/or between two second apertures **333**).

(43) In various embodiments, each complementary baffle plate may comprise a complementary coupling hole configured to receive and/or engage with the coupling rod. For example, complementary baffle plate **300B** may comprise a complementary coupling hole **367** having a complementary shape that is complementary to a cross-sectional shape of coupling rod **450**. Therefore, that coupling rod **450** may be inserted through complementary coupling hole **367**, and complementary coupling hole **367** may engage with coupling rod **450**.

(44) In various embodiments, the complementary coupling hole of a complementary baffle plate may comprise a non-circular shape, such that the coupling rod may engage with the complementary coupling hole and maintain the complementary baffle plate a desired position (e.g., so complementary baffle plate **300B** does not rotate about coupling rod **450** within trap housing **103**). In various embodiments, the complementary coupling hole of a complementary baffle plate may comprise a complementary shape that is symmetrical about only one line passing through the coupling hole (e.g., through a center of the complementary coupling hole). That way, the complementary coupling hole may only engage with the coupling rod in a way that disposes the complementary baffle plate in a desired orientation (a self-aligning feature). In various embodiments, to aid in disposing a complementary baffle plate in a desired orientation about the coupling rod, the complementary coupling hole may comprise a complementary reference point which is oriented in a specific complementary angle, and/or a specific position relative to the complementary aperture(s) of the complementary baffle plate. For example, complementary coupling hole **367** may comprise complementary reference point **368** which may be oriented at a specific complementary angle (e.g., such that complementary reference point **368** is aligned with a complementary second aperture **363** and/or between two complementary first apertures **361**).

(45) In various embodiments, the reference point of a coupling hole and the complementary reference point of a complementary coupling hole may dispose the baffle plate and the complementary baffle plate in an orientation such that an aperture of a baffle plate may be aligned along an axis with a complementary solid body portion (or radially proximate to space between complementary apertures) of an adjacent complementary baffle plate in the baffle plate order, wherein the axis spans along the baffle plate order. In various embodiments, the reference and the complementary reference point of a complementary coupling hole may dispose the baffle plate and the complementary baffle plate in an orientation such that a complementary aperture of a complementary baffle plate may be aligned along an axis with a solid body portion (or radially proximate a space between apertures) of an adjacent baffle plate in the baffle plate order, wherein the axis spans along the baffle plate order. For example, coupling hole **347** and reference point **348** may dispose baffle plate **300A**, and complementary coupling hole **367** and complementary reference point **368** may dispose complementary baffle plate **300B**, such that apertures **333** are aligned along an axis with space between complementary apertures **363**, and such that complementary apertures **363** are aligned along an axis with space between apertures **333**.

(46) In various embodiments, the baffle plates and complementary baffle plates may be disposed in a certain baffle plate orientation to achieve desired fluid flow therethrough and contamination deposition thereon during contaminant trap system **100** operation. In various embodiments, the rotational position of the baffle plates and complementary baffle plates about a coupling rod in a baffle plate stack may be offset relative to one another (e.g., by the orientation of the coupling hole and reference point, and the orientation of the complementary coupling hole and complementary reference point) such that the apertures of a baffle plate are not in series and/or aligned with the complementary apertures of a complementary baffle plate along an axis spanning the baffle plate

stack. Further, the apertures of a baffle plate may be in series and/or aligned with, along an axis spanning a baffle plate stack, at least a portion of the complementary solid body portion (or portions of the complementary baffle plate body, e.g., between complementary apertures) of an adjacent complementary baffle plate in the baffle plate stack. Further, the complementary apertures of a complementary baffle plate may be in series and/or aligned with, along an axis spanning a baffle plate stack, at least a portion of the solid body portion (or portions of the baffle plate body, e.g., between apertures) of an adjacent baffle plate in the baffle plate stack. In other words, in various embodiments, the reference point of a coupling hole may be aligned with an aperture of the baffle plate, and the complementary reference point of a complementary coupling hole may be aligned with a complementary solid body portion or a space between the complementary apertures of a complementary baffle plate; and/or, the reference point of a coupling hole may be aligned with a solid body portion or space between the apertures of a baffle plate, and the complementary reference point of a complementary coupling hole may be aligned with a complementary aperture of a complementary aperture of a complementary baffle plate. For example, reference point **348** may be aligned with an aperture of baffle plate **300A**, and complementary reference point **368** may be aligned with a complementary solid body portion **365** of complementary baffle plate **300B**. Accordingly, apertures **331** and **333** of baffle plate **300A** may be in series and/or aligned with complementary solid body portion **365** and/or spaces between complementary apertures **361** and/or **363** of complementary baffle plate **300B**, and complementary apertures **361** and/or **363** of complementary baffle plate **300B** may be in series and/or aligned with solid body portion **335** and/or spaces between apertures **331** and/or **333** of baffle plate **300A**.

(47) FIGS. **5A** and **5B** depict, in accordance with additional embodiments, a baffle plate **500A** and a complementary baffle plate **500B**. Baffle plate **500A** may comprise apertures **533** and solid body portions **535**. Baffle plate **500A** may further comprise coupling hole **547** having a reference point **548**. Reference point **548** may be oriented toward an aperture **533**. Apertures **533** may be equidistant about center of baffle plate **500A**.

(48) Complementary baffle plate **500B** may comprise complementary apertures **563** and complementary solid body portions **565**. Complementary baffle plate **500B** may further comprise complementary coupling hole **567** having a complementary reference point **568**. Complementary reference point **568** may be oriented toward a complementary solid body portion **565**. Complementary solid body portions **565** may be equidistant about center of complementary baffle plate **500B**.

(49) The coupling rod to which baffle plate **500A** and complementary baffle plate **500B** may couple may comprise a cross-sectional shape complementary to coupling hole **547** and complementary coupling hole **567**. That is, the coupling rod may comprise a body and a protrusion complementary to reference point **548** and complementary reference point **568**. The shape and orientation of coupling hole **547** and complementary coupling hole **567**, and reference point **548** and complementary reference point **568**, respectively, may offset the rotational position of the baffle plates and complementary baffle plates about a coupling rod in a baffle plate stack relative to one another. Accordingly, apertures **533** of baffle plate **500A** may be in series and/or aligned with, along an axis spanning a baffle plate stack, complementary solid body portions **565** and/or spaces between complementary apertures **563** of complementary baffle plate **500B**, and complementary apertures **563** of complementary baffle plate **500B** may be in series and/or aligned with, along an axis spanning a baffle plate stack, solid body portion **535** and/or spaces between apertures **533** of baffle plate **500A**.

(50) FIGS. **6A** and **6B** depict, in accordance with further embodiments, a baffle plate **600A** and a complementary baffle plate **600B**. Baffle plate **600A** may comprise apertures **633** and solid body portions **635**. Baffle plate **600A** may further comprise coupling hole **647** having a reference point **648**. Reference point **648** may be oriented toward a solid body portion **635** and/or space between apertures **633**. Apertures **633** may be equidistant about center of baffle plate **600A**.

(51) Complementary baffle plate **600B** may comprise complementary apertures **663** and complementary solid body portions **665**. Complementary baffle plate **600B** may further comprise complementary coupling hole **667** having a complementary reference point **668**. Complementary reference point **668** may be oriented toward a complementary aperture **663**. Complementary apertures **663** may be equidistant about center of complementary baffle plate **600B**.

(52) The coupling rod to which baffle plate **600A** and complementary baffle plate **600B** may couple may comprise a cross-sectional shape complementary to coupling hole **647** and complementary coupling hole **667**. That is, the coupling rod may comprise a body and a protrusion complementary to reference point **648** and complementary reference point **668**. The shape and orientation of coupling hole **647** and complementary coupling hole **667**, and reference point **648** and complementary reference point **668**, respectively, may offset the rotational position of the baffle plates and complementary baffle plates about a coupling rod in a baffle plate stack relative to one another. Reference point **648** may be aligned with a solid body portion **635**, and/or space between apertures **633**, of baffle plate **600A**, and complementary reference point **668** may be aligned with a complementary aperture **663** of complementary baffle plate **600B**. Accordingly, apertures **633** of baffle plate **600A** may be in series and/or aligned with, along an axis spanning a baffle plate stack, complementary solid body portions **665** and/or spaces between complementary apertures **663** of complementary baffle plate **600B**, and complementary apertures **663** of complementary baffle plate **600B** may be in series and/or aligned with, along an axis spanning a baffle plate stack, solid body portion **635** and/or spaces between apertures **633** of baffle plate **600A**.

(53) FIGS. 7A and 7B depict, in accordance with various embodiments, a baffle plate **700A** and a complementary baffle plate **700B**. Baffle plate **700A** may comprise apertures **733** and solid body portions **735**. Baffle plate **700A** may further comprise coupling hole **747** having a reference point **748**. Reference point **748** may be oriented toward a solid body portion **735** and/or space between apertures **733**. Apertures **733** may be equidistant about center of baffle plate **700A**.

(54) Complementary baffle plate **700B** may comprise complementary apertures **763** and complementary solid body portions **765**. Complementary baffle plate **700B** may further comprise complementary coupling hole **767** having a complementary reference point **768**. Complementary reference point **768** may be oriented toward a complementary aperture **763**. Complementary apertures **763** may be equidistant about center of complementary baffle plate **700B**.

(55) The coupling rod to which baffle plate **700A** and complementary baffle plate **700B** may couple may comprise a cross-sectional shape complementary to coupling hole **747** and complementary coupling hole **767**. That is, the coupling rod may comprise a body and a protrusion complementary to reference point **748** and complementary reference point **768**. The shape and orientation of coupling hole **747** and complementary coupling hole **767**, and reference point **748** and complementary reference point **768**, respectively, may offset the rotational position of the baffle plates and complementary baffle plates about a coupling rod in a baffle plate stack relative to one another. Reference point **748** may be aligned with a solid body portion **735**, or space between apertures **733**, of baffle plate **700A**, and complementary reference point **768** may be aligned with a complementary aperture **763** of complementary baffle plate **700B**. Accordingly, apertures **733** of baffle plate **700A** may be in series and/or aligned with, along an axis spanning a baffle plate stack, complementary solid body portions **765** and/or spaces between complementary apertures **763** of complementary baffle plate **700B**, and complementary apertures **763** of complementary baffle plate **700B** may be in series and/or aligned with, along an axis spanning a baffle plate stack, solid body portion **735** and/or spaces between apertures **733** of baffle plate **700A**.

(56) Any of the pairs of baffle plates and complementary baffle plates discussed herein (or individual plates) may be input into a baffle plate stack (e.g., to replace baffle plates **300A** and complementary baffle plates **300B** in baffle plate stack **400B**).

(57) In various embodiments, between each baffle plate and complementary baffle plate in a baffle plate stack, there may be a spacer configured to space the adjacent baffle plates and complementary

baffle plates. For example, with reference to FIG. 4B, baffle plates **300A** and complementary baffle plates **300B** may be separated by spacers **303** (an example of spacers **133** in FIG. 2). A spacer may be disposed between every plate in a baffle stack (e.g., between baffle plates and complementary baffle plates, between end plates and baffle plates and/or complementary baffle plates, or the like), to achieve any desired spacing between two plates. Such spacing may achieve a desired pressure drop in the fluid airflow flowing through trap housing **103** and the apertures and complementary apertures in the baffle plates and complementary baffle plates comprised therein.

(58) In various embodiments, a baffle plate stack may comprise at least one end plate disposed adjacent to a first and/or last baffle plate (or complementary baffle plate) in the baffle plate order. An end plate may have an end plate coupling hole similar to the coupling hole of a baffle plate and the complementary coupling hole of a complementary baffle plate, configured to engage with the coupling rod. An end plate may further comprise at least one end plate aperture disposed through an end plate body between a first and second surface of the end plate. For example, as shown in FIG. 4A, end plate **410** may comprise end plate apertures **412**. End plate apertures may be disposed through an end plate in any suitable design or arrangement. In various embodiments, a portion of the end plate that does not comprise an aperture may be an end plate solid body portion (e.g., end plate solid body portion **414**).

(59) In various embodiments, an end plate (e.g., end plate **410** in FIG. 4A) may be configured to be disposed adjacent to the internal surface of the first end or second end of trap housing **103**, such that the outer surface of the end plate may be adjacent to and/or in contact with the internal surface of trap housing **103**. Such a configuration may allow greater heat conductance into the baffle plate stack, e.g., from an external heat source, such as a heater jacket (e.g., heater jacket **800** depicted in FIG. 8) configured to be coupled around contaminant trap system **100** and/or trap housing **103**. In various embodiments, an end plate (e.g., end plates **420** in FIG. 4B) may be configured to be spaced from the internal surface of the first end or second end of trap housing **103**, such that there is a space between the outer surface of the end plate and the internal surface of trap housing **103**. The space between the internal surface of trap housing **103** and an endplate may be achieved by an end plate comprising flanges (e.g., flanges **424**), or a spacer disposed therebetween. Such a configuration may achieve a desired pressure drop of the fluid flow through trap housing **103**, and/or provide greater area for contaminant deposition within trap housing **103** and the baffle plate stack (e.g., baffle plate stack **400B**).

(60) In various embodiments, an end plate may comprise an end plate aperture and/or end plate aperture arrangement that causes an end plate aperture to be in series and/or aligned with (e.g., along an axis spanning a baffle plate stack) an aperture disposed through the next adjacent plate in the baffle plate stack. For example, end plate apertures **422** of end plate **420** may be in series and/or aligned with, along an axis spanning a baffle plate stack, apertures **331** and/or **333** of baffle plate **300A**. That way, fluid entering and flowing through trap housing **103** and baffle plate stack **400B** will deposit less contaminants on plates more proximate to fluid inlet **101A**, thus decreasing the risk of outgassing of contamination from contaminant trap system **100** to upstream components such as a reaction chamber.

(61) In various embodiments, the plates in a baffle stack, including baffle plates, complementary baffle plates, and end plates, may be coupled to the coupling rod, and secured by a fastener. For example, fasteners **402** (e.g., a screw, nail, clamp, or the like) may engage with coupling rod **450** (e.g., via threading, force, and/or the like), and secure baffle plates **300A**, complementary baffle plates **300B**, end plates **420** and/or spacers **303**.

(62) In various embodiments, fastener **402** may be disposed in and/or coupled to a sleeve **407**, which may be disposed in an end of coupling rod **450**. Sleeve **407** may be configured to provide a buffer between fastener **402** and the adjacent surface of coupling rod **450** to avoid galling.

(63) In various embodiments, one or more plates in the baffle plate stack may comprise an indicator to readily convey to a user or assembler of the baffle plate stack which plate is disposed in

which baffle plate stack position. Therefore, in various embodiments, for example, baffle plates **300A** may comprise indicator **304** (e.g., a notch) to readily indicate that the notched or otherwise marked plate is a baffle plate **300A**. Accordingly, a user or assembler of the baffle plate stack may be able to readily discern whether the correct order of baffle plates and complementary baffle plates is achieved. Any of the plates of the baffle plate stacks discussed herein may comprise an indicator.

(64) In various embodiments, a baffle plate stack may be palindromic, such that the order of components is the same from either end of the baffle plate stack. As illustrated in FIG. **4B**, baffle plate stack **400B** begins and ends with an end plate **420**, and therebetween, an odd number of baffle plates **300A** alternating in a baffle plate order with an even number of complementary baffle plates **300B** such that the baffle plate order starts and ends with a baffle plate **300A**. Accordingly, someone assembling a contaminant trap system may insert baffle plate stack **400B** into trap housing **103** without worry about whether baffle plate stack **400B** is right side up or upside down.

(65) In various embodiments, any surface of a baffle plate stack or other contaminant trap system component that will interact with fluid flowing through a contaminant trap system may receive contaminant deposit (which is the objective of the methods and systems discussed herein, in order to remove the contaminant from the fluid to avoid contamination of downstream reactor system components). Therefore, to increase available surface area of the components, the surfaces may be textured (e.g., by bead blasting). For example, the surfaces of the baffle plates and complementary baffle plates (including the outer edges thereof), spacers, internal wall of the trap housing, the edges of the apertures and complementary apertures, and/or any other surface may be textured.

(66) In various embodiments, the components of contaminant trap system **100** may be clamped and/or sealed together by clamping ring **144**. A clamping ring **144** may be disposed around upper housing **103A** and/or lower housing **103B**, and may be configured to be tightened to hold components of contaminant trap system **100** together.

(67) In various embodiments, a trap structure comprised in a contaminant trap system may comprise structures for trapping contaminants other than a baffle plate stack, discussed above. For example, with reference to FIGS. **10A** and **10B**, a contaminant trap system may comprise a trap structure **1000** disposed in the trap housing (e.g., trap housing **103** shown in FIG. **2**) comprising a plurality of rods **1055**. Rods **1055** may be arranged in an arrangement **1050** to direct fluid flowing between rods **1055** along a desired path. Rods **1055** may span between components which may provide stability for rods **1055** within trap structure **1000**. For example, rods **1055** may be coupled to and/or span between a baffle plate **1010** and a base plate **1020**. Rods **1055** may be substantially perpendicular to the baffle plate **1010** and/or base plate **1020**, and/or substantially parallel to an axis spanning between the fluid inlet **101A** and fluid outlet **101B** of the trap housing **103** (shown in FIG. **2**) (as used in this context, the term “substantially” means plus or minus 20 degrees from perpendicular or parallel, respectively). In various embodiments, the rods in the trap structure may be integral or monolithic with the baffle plate and/or base plate.

(68) In various embodiments, as depicted in FIGS. **10A** and **10B**, baffle plate **1010** may comprise recesses **1014** disposed into an inner side **1011** of baffle plate **1010**. Recesses **1014** may comprise a shape that is complementary to the cross-sectional shape of a respective rod **1055**. A first end **1052** of each rod **1055** may be disposed into a respective recess **1014**, thus coupling rods **1055** to baffle plate **1010**. Similarly, in various embodiments, base plate **1020** may comprise recesses **1024** disposed into an inner side **1021** of base plate **1020** (inner side **1021** of base plate **1020** may face baffle plate **1010**). Recesses **1024** may comprise a shape that is complementary to the cross-sectional shape of a respective rod **1055**. A second end **1054** of each rod **1055** may be disposed into a respective recess **1024**, thus coupling rods **1055** to base plate **1020**. The rods of the trap structure may be coupled to a baffle plate and/or base plate by the rods resting in the respective recesses in the baffle plate and/or base plate, by tight fit within the respective recesses, threading on the base plate, baffle plate, and the rod ends to allow the rods to screw into the base plate and/or baffle plate, or the like.

(69) In various embodiments, the rods may be coupled to a baffle plate and/or a base plate, whether or not the baffle plate and/or the base plate have recesses configured to receive the rods, in any suitable manner, for example, via welding, tightening between the baffle and base plates, adhesive, or the like.

(70) In various embodiments, the trap structure may comprise a center support (e.g., center support **1025**), which may be configured to couple two or more components of the trap structure. For example, center support **1025** of trap structure **1000** may couple baffle plate **1010** to base plate **1020**, with rods **1055** disposed therebetween. Center support **1025** may be disposed through a support hole **1016** in baffle plate **1010** configured to receive center support **1025** therethrough. The shape of support hole **1016** may be complementary to the cross-sectional shape of center support **1025**. Center support **1025** may be coupled and/or secured to baffle plate **1010** by a fastener (e.g., nut **1002** and/or seal **1004**) disposed around center support **1025** and in contact with baffle plate **1010**. In various embodiments, the fastener may comprise threading complementary to threading on the center support, such that the fastener is threaded onto the center support, and then tightened toward the base plate to push the baffle plate and base plate together. Thus, in various embodiments, rods **1055** disposed between baffle plate **1010** and base plate **1020** may be held in place by the force between baffle plate **1010** and base plate **1020** from center support **1025** and fastener **1002**. The center support may be a separate component, or may be integral or monolithic with the baffle plate and/or base plate of the trap structure.

(71) In various embodiments, rods **1055** may be disposed around (i.e., about) a center area of the base plate (e.g., the portion at or proximate center support **1025** of base plate **1020**). The center area may not comprise any rods. The center area may comprise one or more flow holes (e.g., flow holes **1027**) disposed through the base plate through which fluid flowing through the contaminant trap system and trap structure may flow. Thus, with airflow going through the trap housing (e.g., caused by vacuum pressure from vacuum pump **28**, shown in FIG. **1**), fluid flowing through the trap housing (including lower housing **103B**) may be required to flow through arrangement **1050** of rods **1055**, contacting rods **1055** while so doing, before exiting trap structure **1000** through flow holes **1027**, and exiting the trap housing through fluid outlet **101B** of the trap housing. The flow holes may be aligned and/or misaligned with fluid outlet **101B**.

(72) In various embodiments, rods in the trap structure may be disposed in any suitable arrangement. For example, rods **1055** may be disposed spaced apart (i.e., not contacting one another), or may be contacting one another, such that fluid may flow between rods **1055**. The spacing of the rods may provide a convoluted path for fluid flowing through the trap structure, thus increasing the chances that the fluid will contact more surfaces and contaminants in the fluid will be deposited onto such surfaces within the trap. The rods may comprise any suitable shape or length. For example, the rods may comprise a circular cross-sectional shape (such as those shown in FIGS. **10A** and **10B**), or the rods may comprise, for example, a hexagonal, octagonal, triangular, or square cross-sectional shape, or any other suitable cross-sectional shape. As another example, the rods may have a cross-sectional length (e.g., the diameter of a circle) of approximately 2 millimeters (mm) (“approximately” as used in this context means plus or minus 0.5 mm). As another example, the rods may have a length (e.g., the distances spanning between the baffle plate and base plate) of approximately 20 centimeters (cm) (“approximately” as used in this context means plus or minus 5 cm). The rods may comprise a high surface-area-to-volume ratio, for example, a surface-area-to-volume ratio of at least 50:1, at least 100:1, at least 150:1, or at least 200:1. In various embodiments, the rods may comprise a textured outer surface, threading along the rods, or any other structure configured to increase the outer surface area of the rods for contaminant deposition thereon.

(73) The rods in a trap structure may comprise any suitable material, such as steel, aluminum, or any other metal or alloy thereof, ceramic material, or the like.

(74) In various embodiments, the base plate of a trap structure (e.g., base plate **1020**) may be

disposed in the trap housing and support other components of the trap structure. In various embodiments, an outer side of base plate **1020** (opposite of inner side **1021**) may be disposed spaced from a housing bottom surface (housing bottom surface **102**) of the trap housing. To support the base plate being spaced from the trap housing bottom surface, the trap housing may comprise a support (e.g., support **1006**) protruding from the trap housing to hold the base plate in place. For example, support **1006** may protrude from the interior wall of the trap housing to support base plate **1020** in place, spaced from bottom surface **102** of the trap housing. In various embodiments, a support may protrude from another surface of the trap housing, for example, from the bottom surface, to hold the base plate in place. In various embodiments, the base plate outer surface may be disposed against or adjacent to the trap housing bottom surface.

(75) In various embodiments, the baffle plate of a trap structure (e.g., baffle plate **1010**) may cause fluid flow entering the trap housing to take a certain path (e.g., a path that will increase fluid flow around and in contact with rods **1055**, and/or increase removal of contaminants from the fluid). Baffle plate **1010** may reduce or prevent fluid flow from traveling around first ends **1052** of rods **1055**. That is, baffle plate **1010** may form at least a partial seal between baffle plate **1010** and the first ends **1052** of rods **1055**. In various embodiments, the shape of baffle plate **1010** may be smaller than a cross-sectional shape of the trap housing such that baffle plate edge **1012** does not contact the interior wall of the trap housing. Thus, there may be a space between the baffle plate edge and the interior wall of the trap housing, and/or between the rods **1055** and the interior wall of the trap housing (e.g., space **1075** between the interior wall of lower housing **103B** and baffle plate edge **1012** and/or rods **1055**). Baffle plate **1010** may be configured to cause at least a portion of fluid flow within the trap housing to flow around baffle plate edge **1012** toward and through arrangement **1050** of rods **1055** (e.g., through space **1075**), and toward flow holes **1027**.

(76) In various embodiments, the base plate may form at least a partial seal with the interior wall of the trap housing. For example, the outer edge of base plate **1020** may be disposed against or adjacent to the interior wall of lower housing **103B**, such that little or no fluid may pass therebetween. Therefore, the fluid flowing through trap structure **1000** may be directed around baffle plate **1010** (and/or through a baffle plate comprising holes disposed therethrough) to flow through the arrangement **1050** of rods **1055**, and exit trap structure **1000** through flow holes **1027**. Thus, contaminants in the fluid may deposit on the surfaces of the trap structure (e.g., the outer surfaces **1053** of rods **1055**, baffle plate **1010**, base plate **1020**, etc.) with little or no fluid flow between base plate **1020** and the interior wall of the trap housing.

(77) The arrangement of components of trap structure **1000** may allow greater heat conductance therethrough. Heating a trap structure may allow for increased growth rates of contaminant films on the trap system components, and improved trapped contaminant film properties such as increased density and decreased flaking. Thermal energy may readily travel through the base plate, rods, and/or baffle plate, whether the thermal energy is provided externally and/or internally. In various embodiments, trap structure **1000** may be heated externally, for example by a heater jacket (e.g., heater jacket **800** depicted in FIG. **8**) being coupled around the contaminant trap system and/or trap housing comprising trap structure **1000**. In various embodiments, trap structure **1000** may be heated internally, for example by a heater (e.g., heater **1026** depicted in FIG. **10B**) disposed in, or coupled to, a component of trap structure **1000** (e.g., in base plate **1020** and/or center support **1025** comprising a heater **1026**). Especially in embodiments in which rods **1055** comprise a metal material, such as steel or aluminum (or alloys thereof), thermal energy would readily travel between base plate **1020** (receiving thermal energy from heater **1026** and/or from a heater jacket through the trap housing), rods **1055**, and baffle plate **1010**.

(78) In various embodiments, a trap structure such as trap structure **1000** comprising rods **1055** disposed between a baffle plate **1010** and a base plate **1020**, in addition to providing plentiful surface area on which contaminants may deposit, may also have the benefit of reusability and easy maintenance. In response to trap structure **1000** being used and/or saturated with contaminants, the

components of trap structure **1000** (e.g., rods **1055**, baffle plate **1010**, and base plate **1020**) may be disassembled (and/or removed from the trap housing), easily cleaned, and then reassembled for subsequent use. The trap structure may be disassembled, for example, by uncoupling the fastener **1002** from center support **1025**. If one or more of the components are damaged or otherwise need replacement, such a replacement can easily be completed. Other pre-existing components for trap structures are one-time use items and/or difficult to clean.

(79) In various embodiments, a trap structure comprised in a contaminant trap system may comprise a plurality of tubes through which a fluid may flow. Each tube may comprise a bore (e.g., bores **1157**) disposed therethrough for the length of the tube, allowing contaminants to deposit on inner and outer surfaces of the tubes. For example, with reference to FIG. **11**, trap structure **1100** may comprise an arrangement **1150** of tubes **1155**. Trap structure **1100** may be disposed in a trap housing (e.g., trap housing **103** shown in FIG. **2**) such that tubes **1155** span at least partially between a trap housing top surface and bottom surface (e.g., along the direction of fluid flow through the trap housing, and/or substantially parallel to an axis spanning between the fluid inlet **101A** and fluid outlet **101B** of the trap housing **103** (shown in FIG. **2**) (as used in this context, the term “substantially” means plus or minus 20 degrees from parallel)). The arrangement of tubes in a trap structure may be complementary to the shape of the trap housing, such that the tubes on the outer perimeter of the tube arrangement may abut or be disposed adjacent to the interior wall of the trap housing. For example, arrangement **1150** of tubes **1155** may be configured to be disposed in a hexagonal trap housing. In various embodiments, the tubes of a tube arrangement for a trap structure may comprise a circular arrangement configured to be disposed into a circular trap housing (e.g., into lower trap housing **103B** shown in FIG. **10A**).

(80) The tubes may be arranged in any suitable manner relative to one another. The tube arrangement may be configured to limit or minimize the space between the tubes. For example, as shown in FIG. **11**, in accordance with various embodiments, tubes **1155** may be packed hexagonally, such that each tube **1155** (except tubes on the outer perimeter) may be surrounded by six tubes **1155**. Therefore, each tube **1155** (except tubes on the outer perimeter) may abut or be in contact with six other tubes **1155**. This hexagonal packing allows uniform packing of tubes **1155** and limits the space therebetween, providing dense packing with circular tubes. Such dense packing prevents shifting of the tubes relative to one another. Also, the hexagonal packing of tubes forms triangular spaces (e.g., spaces **1159**) having concave sides between the contacting tubes. These spaces between tubes allow for additional space for fluid to flow through and additional surface area (on the outside of the tubes) upon which contaminants may deposit. The hexagonal packing of tubes does not necessarily apply to the outer shape of the tube arrangement, and may be implemented in a tube arrangement having a circular outer shape.

(81) The tubes in a trap structure may comprise any suitable shape or dimensions. In various embodiments, the tubes may comprise a circular cross-sectional outer shape (e.g., tubes **1155**), or any other suitable cross-sectional shape configured to allow a desired arrangement of the tubes. In various embodiments, the tube bores may comprise a circular cross-sectional shape (e.g., bores **1157**), or any other suitable cross-sectional bore shape. In various embodiments, the tubes may have a cross-sectional length (e.g., an outer diameter of tubes **1155**) of approximately 2 millimeters (mm). In various embodiments, the tubes may have an inner diameter (e.g., the length across the bores, such as the diameter of bores **1157**) of approximately 1 mm (“approximately” as used in this context means plus or minus 0.5 mm). In various embodiments, the tubes may have a length of approximately 20 centimeters (cm) (“approximately” as used in this context means plus or minus 5 cm). The tubes may comprise a high surface-area-to-volume ratio, for example, a surface-area-to-volume ratio of at least 50:1, at least 100:1, at least 150:1, or at least 200:1. For example, tubes in a hexagonal packing arrangement that are approximately 20 cm in length, having an outer diameter of about 2 mm and an inner diameter of 1 mm, filling a trap housing having a diameter of about 19 cm, provide significant surface area to receive contaminant deposition. The surface area of the tube

bore in such an example would provide approximately six square meters of trapping surface, and the gaps between the tubes (e.g., spaces **1159**) would provide slightly less than six square meters, for a total surface area of about twelve square meters. Assuming a typical deposition process in a reactor produces three square micrometers of contaminant deposit within a trap, the surface area provided by the trap structures comprising the tubes in the arrangement and dimensions discussed would allow the same trap structure to be used for numerous deposition cycles before needing maintenance or replacement.

(82) In various embodiments, the outer and/or inner surfaces of the tubes may comprise a textured outer surface, threading along the outer and/or inner surfaces, or any other structure configured to increase the outer surface area of the tubes for contaminant deposition thereon.

(83) In various embodiments, the tubes may be coupled in any suitable manner such as adhesive, welding, and/or tight fit within the trap housing. As shown in FIG. **11**, tubes **1155** are coupled together to maintain arrangement **1150** by tensioning device **1188**, which may be a clamping ring (similar to clamping ring **114**), a belt, an elastic band, or the like.

(84) In various embodiments, an arrangement **1150** of tubes **1155** may comprise at least one support **1125**. A support **1125** may be a rod or other structure that protrudes at least outwardly from the bottom of arrangement **1150** (i.e., a support **1125** extends closer to a bottom surface of a trap housing than tubes **1155**). In various embodiments, an arrangement of tubes may comprise more than one support (e.g., three supports **1125**, as shown in FIG. **11**). Supports **1125** may be configured to support the arrangement **1150** of tubes **1155** such that there is a space between the bottom of tubes **1155** and a bottom surface of the trap housing (e.g., bottom surface **102** if trap structure **1100** were disposed in lower trap housing **103B**, shown in FIG. **10A**). Similarly, when disposed in a trap housing, there may be a space between the tops of tubes **1155** and a top surface of the trap housing. For example, the tube arrangement **1150** may simply rest in a position within the trap housing that causes a space between the tops of tubes **1155** and a top surface of the trap housing (e.g., because of the way upper and lower housings of the trap housing fit together). As another example, supports **1125** may also protrude outwardly from the top of the arrangement **1150** (i.e., a support **1125** extends closer to a top surface of a trap housing than tubes **1155**). Thus, if a lid or upper housing of a trap housing is placed on the trap structure, the lid or upper housing will come to rest against the tips of supports **125**, thus allowing a space between the top surface of the trap housing and the tops of tubes **1155**. Such a space would allow fluid flowing into a trap housing (e.g., through fluid inlet **101A**, shown in FIG. **2**) to disperse and utilize more of tubes **1155** to trap contaminants.

(85) In various embodiments, a structure, such as a baffle plate with holes, showerhead, or the like may be disposed above the tube arrangement in a trap housing to disperse fluid flowing thereto in a desired manner to increase utilization of the surface area provided by the tubes.

(86) The arrangement of components of trap structure **1100** may allow greater heat conductance therethrough. Heating a trap structure may allow for increased growth rates of contaminant films on the trap system components, and improved trapped contaminant film properties such as increased density and decreased flaking. Thermal energy may readily travel through the trap housing, supports **1125**, and/or tubes, whether the thermal energy is provided externally and/or internally. In various embodiments, trap structure **1100** may be heated externally, for example by a heater jacket (e.g., heater jacket **800** depicted in FIG. **8**) being coupled around the contaminant trap system and/or trap housing comprising trap structure **1100**. In various embodiments, trap structure **1100** may be heated internally, for example by a heater being disposed in the arrangement **1150** of tubes **1155**. For example, a tube within the tube arrangement (e.g., a tube at or close to the center of the arrangement) may be replaced by a heater, and/or a support **125** may be, or may comprise, a heater. Especially in embodiments in which tubes **1155** comprise a metal material, such as steel or aluminum (or alloys thereof), thermal energy would readily travel through tubes **1155** and/or supports **1125** (e.g., if receiving thermal energy from a heater jacket through the trap housing or from an internal heater).

(87) In various embodiments, a trap structure such as trap structure **1100** comprising tubes **1155**, in addition to providing plentiful surface area on which contaminants may deposit, may also have the benefit of reusability and easy maintenance. In response to trap structure **1100** being used and/or saturated with contaminants, the components of trap structure **1100** (e.g., tubes **1155**, supports **1125**, tensioning device **1188**) may be easily removed from a trap housing and/or disassembled, cleaned, and then reassembled for subsequent use. The trap structure may be disassembled, for example, by uncoupling the tensioning device **1188** from tubes **1155**. If one or more of the components are damaged or otherwise need replacement, such a replacement can easily be completed.

(88) In various embodiments, a trap structure comprised in a contaminant trap system may comprise a corrugated sheet through which fluid may flow and deposit contaminants thereon. With reference to FIG. **12**, in accordance with various embodiments, a corrugated trap structure **1200** may comprise a corrugate sheet **1250** coupled to a noncorrugated sheet **1280**. Spaces **1260** between corrugated sheet **1250** and noncorrugated sheet **1280** may allow fluid to flow therethrough and contaminants to deposit on the surface area provided therein by the sheets. Corrugated trap structure **1200** may be disposed in a trap housing (e.g., trap housing **103** shown in FIG. **2**) such that spaces **1260** span at least partially between a trap housing top surface and bottom surface (e.g., along the direction of fluid flow through the trap housing). Sheets **1250** and **1280** may be spiraled in any suitable shape (e.g., in a circular shape, such as that shown in FIG. **12**, or a square, triangular, rectangular, hexagonal, or octagonal shape). The outer shape of the spiraled sheets may be complementary to the shape of the trap housing in which the trap structure will be disposed. For example, corrugated trap structure **1200** may be configured to be disposed in a circular trap housing, such as trap housing **103** shown in FIG. **2**. Thus, the corrugated sheet or noncorrugated sheet may abut or be disposed adjacent to the interior wall of the trap housing. The corrugated and noncorrugated sheets may be spiraled or arranged such that a middle void **1205** may be decreased or minimized to cause fluid flowing therethrough to flow through spaces **1260** as opposed to other paths through corrugated trap structure **1200**.

(89) In various embodiments, corrugated trap structure **1200** may comprise at least one support (e.g., support **1125** shown in FIG. **11**). A support may be a rod or other structure that protrudes outwardly from the bottom and/or top of corrugated trap structure **1200**. Such a support may be configured to support corrugated trap structure **1200** such that there is a space between the bottom and/or top of corrugated trap structure **1200** and a bottom and/or top surface of the trap housing. Therefore, a space may be created between the bottom of corrugated trap structure **1200** and a bottom surface of the trap housing and/or between the top of corrugated trap structure **1200** and a top surface of the trap housing. Such a space would allow fluid flowing into a trap housing (e.g., through fluid inlet **101A**, shown in FIG. **2**) to disperse and utilize (i.e., flow through) more of spaces **1260** to trap contaminants.

(90) In various embodiments, a structure, such as a baffle plate with holes, showerhead, or the like may be disposed above the corrugated trap structure in a trap housing to disperse fluid flowing thereto in a desired manner to increase utilization of the surface area provided for contaminant deposition.

(91) The arrangement of components of corrugated trap structure **1200** may allow greater heat conductance therethrough. Heating a trap structure may allow for increased growth rates of contaminant films on the trap system components, and improved trapped contaminant film properties such as increased density and decreased flaking. Thermal energy may readily travel through corrugated trap structure **1200**, whether the thermal energy is provided externally and/or internally. In various embodiments, corrugated trap structure **1200** may be heated externally, for example by a heater jacket (e.g., heater jacket **800** depicted in FIG. **8**) being coupled around the contaminant trap system and/or trap housing comprising corrugated trap structure **1200**. In various embodiments, corrugated trap structure **1200** may be heated internally, for example by a heater

being disposed through void **1205** or a heater comprised in a support disposed through void **1205**. Especially in embodiments in which corrugated trap structure **1200** comprises a metal material, such as steel or aluminum (or alloys thereof), thermal energy would readily travel through corrugated trap structure **1200** (e.g., if receiving thermal energy from a heater jacket through the trap housing or from an internal heater).

(92) In various embodiments, the contaminant trap systems, and components comprised therein, may comprise no adhesive or other coupling material to couple any components. The absence of an adhesive, epoxy, or other coupling material mitigates the risk of such a coupling material outgassing and traveling to the reaction chamber, acting as a contaminant therein. Additionally, without such a coupling material, components of the systems discussed herein may not be susceptible to degradation at elevated temperatures, for example, greater than 120° C. Therefore, a contaminant trap system (e.g., contaminant trap system **100** in FIG. 2) and trap structures comprised therein may be moved closer to a reaction chamber of a reactor system (e.g., reaction chamber **4** of reactor system **50** in FIG. 1) than a contaminant trap system comprising a coupling material. Accordingly, a reactor system having a contaminant trap system in accordance with the embodiments discussed herein may be more compact and/or have more feasible configurations and special arrangements.

(93) The contaminant trap systems discussed herein are configured to increase the surface area with which a fluid flowing therethrough may contact to allow more opportunity for contaminant deposition on such surface area. Thus, for example, as discussed herein, the apertures of a baffle plate may not be aligned with and/or in series with the complementary apertures of an adjacent complementary baffle plate in the baffle plate stack. As another example, rods (e.g., rods **1055**) may be arranged such that there is a nonlinear path from an outer perimeter of the arrangement of rods to the flow holes (e.g., flow holes **1027**) allowing fluid to exit the trap structure. As yet another example, tubes (e.g., tubes **1155**) and/or spaces (e.g., spaces **1260**) through a corrugated trap structure (e.g., corrugated trap structure **1200**) may allow contaminants within a fluid to deposit on surfaces within the tubes or paths through a corrugated trap structure.

(94) FIG. 9 illustrates a method **900** of flowing fluid through a contaminant trap system in a reaction system, in accordance with various embodiments. With additional reference to FIGS. 2 and 4B, a fluid may flow from a reaction chamber (e.g., reaction chamber **4** in FIG. 1) to a contaminant trap system (e.g., contaminant trap system **100** in FIG. 2) (step **902**). Contaminant trap system **100** may comprise a fluid inlet **101A** and a fluid outlet **101B** of a trap housing **103**. The fluid may flow into contaminant trap system **100** through fluid inlet **101A**. The fluid may comprise materials which the contaminant trap system is configured to remove from the fluid (e.g., contaminants).

(95) In various embodiments, the fluid may flow through the contamination trap structure (step **904**) comprised in the contaminant trap system. The trap structure may comprise any suitable structural arrangement upon which to collect contaminants from the fluid, such as those discussed herein. In various embodiments, the trap structure in the contamination trap may comprise a baffle plate stack **400B** (e.g., an example of baffle plate stack **130** in FIG. 2) in contaminant trap system **100**. Therefore, the fluid may flow through a plurality of baffle plates **300A** alternating positions in a baffle plate order with a plurality of complementary baffle plates **300B**. The fluid may also flow through at least one end plate (e.g., end plate **420**) comprised in the baffle plate stack on either end of the baffle plate stack. In various embodiments, the fluid may flow through trap structures comprising rods, tubes, and/or corrugated and noncorrugated sheets, as discussed herein.

(96) To flow through the baffle plate stack **400B**, the fluid may flow through a first end plate **420** via end plate apertures **422** and/or around the outer edge of end plate **420**. In flowing through the baffle plate order of baffle plate stack **400B**, the fluid may contact the top surface **322** and bottom surface **324** of baffle plates **300A**, the complementary top surface **352** and complementary bottom surface **354** of complementary baffle plates **300B**, and pass through apertures **331** and **333** of baffle plates **300A** and complementary apertures **361** and **363** of complementary baffle plates **300B**.

Apertures **331** and **333** of baffle plates **300A** may be disposed through baffle plates **300A** and aligned with complementary baffle plates **300B** such that apertures **331** and **333** may be aligned with complementary solid body portions **365** of complementary baffle plates **300B**. Therefore, in response to flowing through apertures **331** and **333** of a baffle plate **300A**, the fluid may contact complementary solid body portions **365** of a subsequent complementary baffle plate **300B** in baffle plate stack **400B**. In response to contacting complementary solid body portions **365** of the next complementary baffle plate **300B**, the fluid may flow toward fluid outlet **101B** and through complementary apertures **361** and **363** of such complementary baffle plate **300B**. Complementary apertures **361** and **363** of complementary baffle plates **300B** may be disposed through complementary baffle plates **300B** and aligned with baffle plates **300A** such that complementary apertures **361** and **363** may be aligned with solid body portions **335** of baffle plates **300A**. Therefore, in response to flowing through complementary apertures **361** and **363** of a complementary baffle plate **300B**, the fluid may contact solid body portions **335** of a subsequent baffle plate **300A** in baffle plate stack **400B**. In response to contacting solid body portions **335** of the next baffle plate **300A**, the fluid may flow toward fluid outlet **101B** and through apertures **331** and **333** of such baffle plate **300A**.

(97) The fluid flow will follow this flow pattern through the baffle plate order of baffle plates **300A** and complementary baffle plates **300B**, until the fluid has passed the final plate in the baffle plate order. The fluid may flow through an end plate **420** on a second end of the baffle plate stack **400B**, contacting the surface of such end plate **420**, and flowing through end plate apertures **422**. The fluid, while flowing through baffle plate stack **400B**, may additionally flow between the outer edges of baffle plates **300A** and complementary baffle plates **300B** and the interior wall surface of outer wall **105**, interacting with and contacting those surfaces.

(98) In various embodiments, to flow through a trap structure with rods (e.g., trap structure **1000**), the fluid may contact and flow around baffle plate **1010** into space **1075**. Then, the fluid may travel through the arrangement **1050** of rods **1055**, contacting rods **1055** before exiting trap structure **1000** through flow holes **1027**.

(99) In various embodiments, to flow through a trap structure with tubes (e.g., trap structure **1100**), the fluid may flow through tubes **1155** before exiting trap structure **1100**.

(100) In various embodiments, to flow through a corrugated trap structure (e.g., corrugated trap structure **1200**), the fluid may flow through spaces **1260** before exiting corrugated trap structure **1200**.

(101) In response to the fluid contacting the surfaces described above (e.g., baffle plates, complementary baffle plates, end plates, interior wall surface of outer wall **105**, rods **1055**, tubes **1155**, corrugated and noncorrugated sheets **1250** and **1280**, etc.), contaminants comprised in the fluid may be deposited or collected from the fluid (step **906**) on surfaces in contaminant trap system and the respective trap structure disposed therein. The surfaces and their positions relative to one another in the contaminant trap system provide increased surface area upon which such contaminant deposition may occur. Some of the surfaces may comprise texturing to provide further available surface area.

(102) In various embodiments, the fluid may flow through fluid outlet **101B** and exit from contaminant trap system (step **908**).

(103) The components of the systems discussed herein may be comprised of any suitable material such as metal or metal alloy (e.g., steel, aluminum, aluminum alloy, or the like).

(104) Although exemplary embodiments of the present disclosure are set forth herein, it should be appreciated that the disclosure is not so limited. For example, although reactor and contaminant trap systems are described in connection with various specific configurations, the disclosure is not necessarily limited to these examples. Various modifications, variations, and enhancements of the system and method set forth herein may be made without departing from the spirit and scope of the present disclosure.

(105) The subject matter of the present disclosure includes all novel and nonobvious combinations and subcombinations of the various systems, components, and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. A contaminant trap system of a reactor system, comprising: a trap housing comprising a housing outer wall; a plurality of baffle plates disposed in the trap housing, wherein each baffle plate comprises a design comprising: a first plurality of apertures spanning through a baffle plate body between a top baffle plate surface and a bottom baffle plate surface; and a solid body portion, wherein the design for each baffle plate is the same; a plurality of complementary baffle plates disposed in the trap housing between a first end and a second end of the trap housing, wherein each complementary baffle plate comprises a complementary design comprising: a first plurality of complementary apertures spanning through a complementary baffle plate body between a top complementary baffle plate surface and a bottom complementary baffle plate surface; and a complementary solid body portion, wherein the complementary design for each complementary baffle plate is the same, and wherein the complementary design is different than the design for each baffle plate, wherein the plurality of baffle plates and the plurality of complementary baffle plates are comprised in a baffle plate stack and disposed in a baffle plate order between a first end and a second end of the baffle plate stack in which the plurality of baffle plates alternates with the plurality of complementary baffle plates such that no two baffle plates from the plurality of baffle plates and no two complementary baffle plates from the plurality of complementary baffle plates are adjacent in the baffle plate order, wherein a first baffle plate and a second baffle plate, from the plurality of baffle plates, and a first complementary baffle plate, from the plurality of complementary baffle plates, are disposed in the baffle plate order in the trap housing such that the first baffle plate is adjacent to the first complementary baffle plate and the second baffle plate is adjacent to the first complementary baffle plate, and wherein the first baffle plate and the first complementary baffle plate are disposed in a baffle plate orientation such that at least a portion of a first aperture of the first baffle plate and at least a portion of the complementary solid body portion of the first complementary baffle plate are aligned along a first axis spanning between the first end and the second end of the trap housing, and such that at least a portion of the solid body portion of the first baffle plate and the second baffle plate, and at least a portion of a complementary aperture of the first plurality of complementary apertures of the first complementary baffle plate, are aligned along a second axis spanning between the first end and the second end of the trap housing, wherein the first plurality of apertures of each baffle plate is disposed on a radially inward portion of each baffle plate, and wherein the solid body portion is disposed on a radially outward portion of each baffle plate, wherein the radially outward portion of each baffle plate does not comprise an aperture, and wherein the first plurality of complementary apertures of each complementary baffle plate is disposed on a radially outward portion of each complementary baffle plate, and wherein the complementary solid body portion is disposed on a radially inward portion of each complementary baffle plate, wherein the radially inward portion of each complementary baffle plate does not comprise a complementary aperture.

2. The contaminant trap system of claim 1, further comprising a coupling rod disposed in the trap housing and spanning between the first end and the second end of the trap housing, wherein the first baffle plate comprises a first coupling hole disposed through the first baffle plate body, wherein the coupling rod is disposed through the first coupling hole, and wherein the first complementary baffle plate comprises a first complementary coupling hole disposed through the first complementary baffle plate body, wherein the coupling rod is disposed through the first complementary coupling hole.

3. The contaminant trap system of claim 2, wherein the coupling rod comprises a noncircular cross-

- section, wherein the first coupling hole of the first baffle plate and the first complementary coupling hole of the first complementary baffle plate each comprise a shape complementary to the noncircular cross-section of the coupling rod.
4. The contaminant trap system of claim 3, wherein a reference point of the first coupling hole is disposed in a first orientation, and a complementary reference point of the first complementary coupling hole is disposed in a first complementary orientation, wherein the first orientation and the first complementary orientation dispose the first baffle plate and the first complementary baffle plate about the coupling rod to achieve the baffle plate orientation.
 5. The contaminant trap system of claim 4, further comprising a spacer between the first baffle plate and the first complementary baffle plate to provide a space therebetween.
 6. The contaminant trap system of claim 1, wherein the baffle plate stack further comprises an end plate disposed such that, at least one of: the first baffle plate is between the end plate and the first complementary baffle plate, or the first complementary baffle plate is between the end plate and the first baffle plate, wherein the end plate comprises an end plate aperture and an end plate solid body portion.
 7. The contaminant trap system of claim 6, wherein the housing outer wall of the trap housing comprises an interior wall surface, wherein an outer edge of at least one of the first baffle plate and the first complementary baffle plate is disposed adjacent to the interior wall surface.
 8. The contaminant trap system of claim 7, wherein at least one of the top baffle plate surface, the bottom baffle plate surface, the top complementary baffle plate surface, the bottom complementary baffle plate surface, the outer edge of at least one of the first baffle plate and the first complementary baffle plate, and the interior wall surface is textured.
 9. The contaminant trap system of claim 6, further comprising a heater jacket coupled to the trap housing.
 10. The containment trap system of claim 1, wherein the baffle plate order is palindromic between the first end and the second end of the baffle plate stack.
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